AI for Anomaly Detection

Lecture 5: Novelty detection with residual models (part I)





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#	Date	Topic
1	15.09	Introducation to anomaly detection (AD)
2	22.09	Supervised ML for AD
3	29.09	Unsupervied ML for AD I
4	06.10	Unsupervised ML for AD II
5	13.10	Residual models: reconstruction ML methods
6	20.10	Residual models: reconstruction neural networks
7	27.10	Residual models: regression models
8	03.11	Initial assignment presentations – obligatory presence in person
9	10.11	Residual models: advanced regression neural networks (CNNs, TCNs)
10	17.11	Domain shifts and data scarcity: transfer learning
11	24.11	Contamination and mislabeling
12	01.12	Quantifying uncertainty, physics informed ML
13	08.12	Recent advances in AD research
14	15.12	Final assignment presentations – obligatory presence in person

Different approaches to anomaly detection

- *Supervised methods:* learn to represent normal and abnormal data based on previous examples.

  ➔ need labeled examples of normal and abnormal.

  👍 high accuracy, easier to validate, anomaly class is clear.

  👎 Disadvantage: high imbalance, high requirements from the data, novelties cannot be discovered.

- *Unsupervised methods:* detect patterns in the data that correspond to normal/ abnormal behaviour without any labels.

  ➔ no labels needed.

  👍 Low requirements on data.

  👎 Harder to achieve high performance, hard to evaluate.

- *Novelty detection (most common):* learn to represent normal data and compare unknown patterns to the representation.

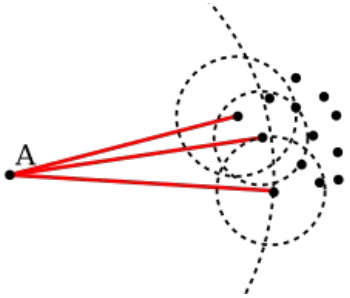
  ➔ need labels only for normal examples.

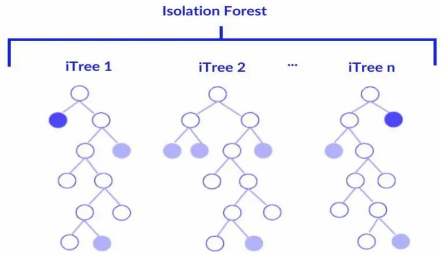
  👍 normal data is usually abundant, detection independent of anomaly type.

  👎 Hard to validate, anomaly class unclear.

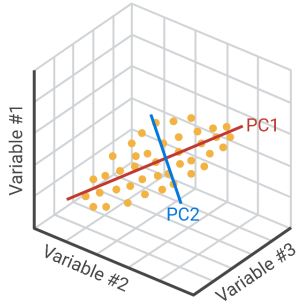
Recap

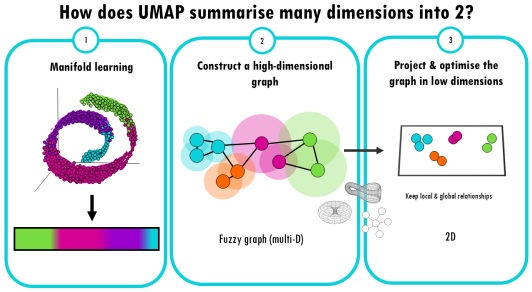
- Unsupervised AD algorithms
  - Hotelling's T^2
  - DBSCAN clustering
  - OCSVM
  - Local Outlier Factor
  - Isolation forest

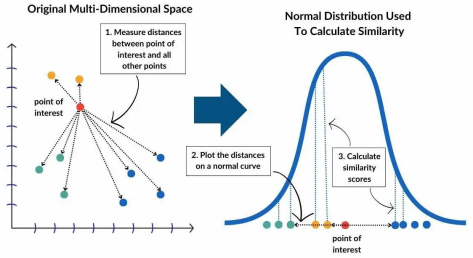




- Unsupervised dimension reduction
  - PCA
  - t-SNE
  - UMAP



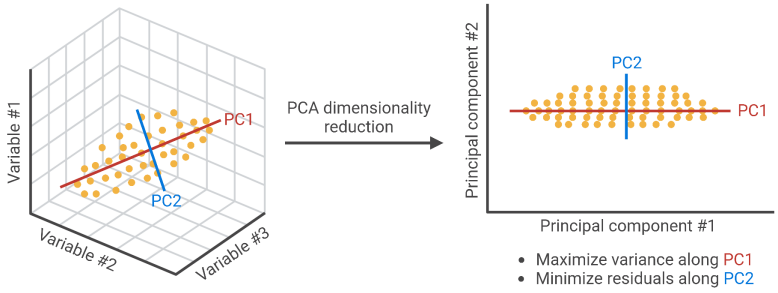




Today: novelty detection with reconstruction methods

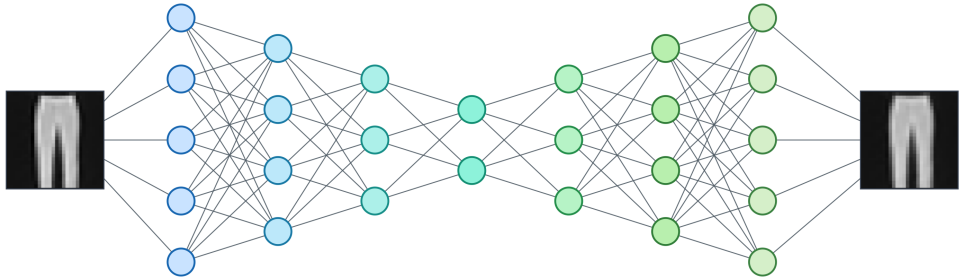
I. Reconstruction with PCA

- Theory
- Example: machine sensor data
- Example + Exercise: image anomaly detection



II. Reconstruction with neural networks

- Introduction to NNs
- Autoencoder NNs
- Example: image anomaly detection



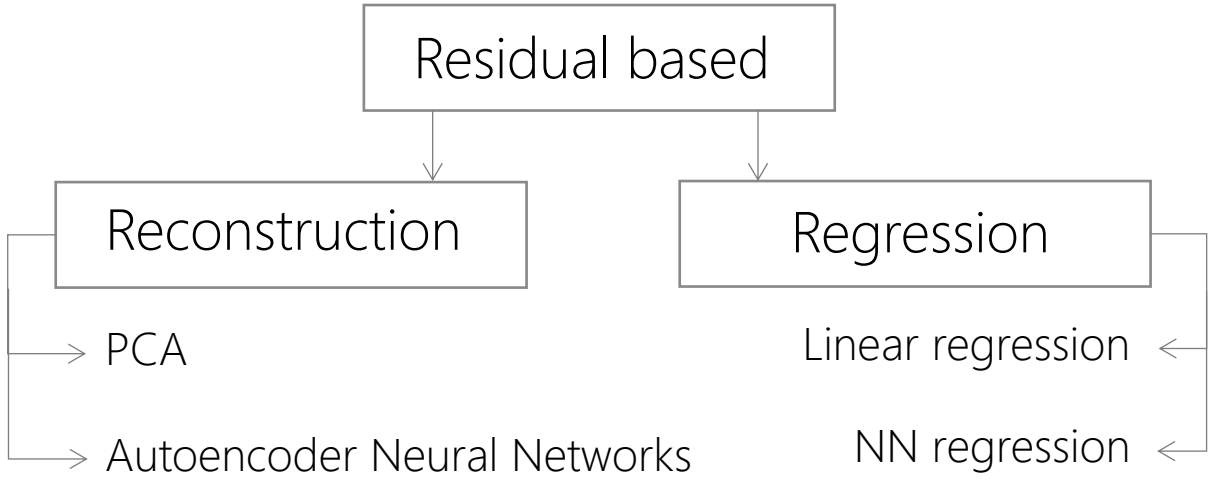
Novelty detection methods

Common situation: only normal data is available.

- Anomalies are rare/ costly/ risky.
- Not all expected types of anomalies have been observed.

➔Residual based approaches:

- train with clean (normal) data.
- Use residuals as anomaly scores.



AD with Principal Component Analysis (PCA)

- Reduce the dimension of the data to a lower dimensional space.
- Reconstruct the data from the reduced space.
- Use only normal data to derive the PCA transformation.

➔Reconstructing abnormal samples will be unsuccessful, assuming they are generated by a different process.

➔Reconstruction errors can be used as anomaly scores.

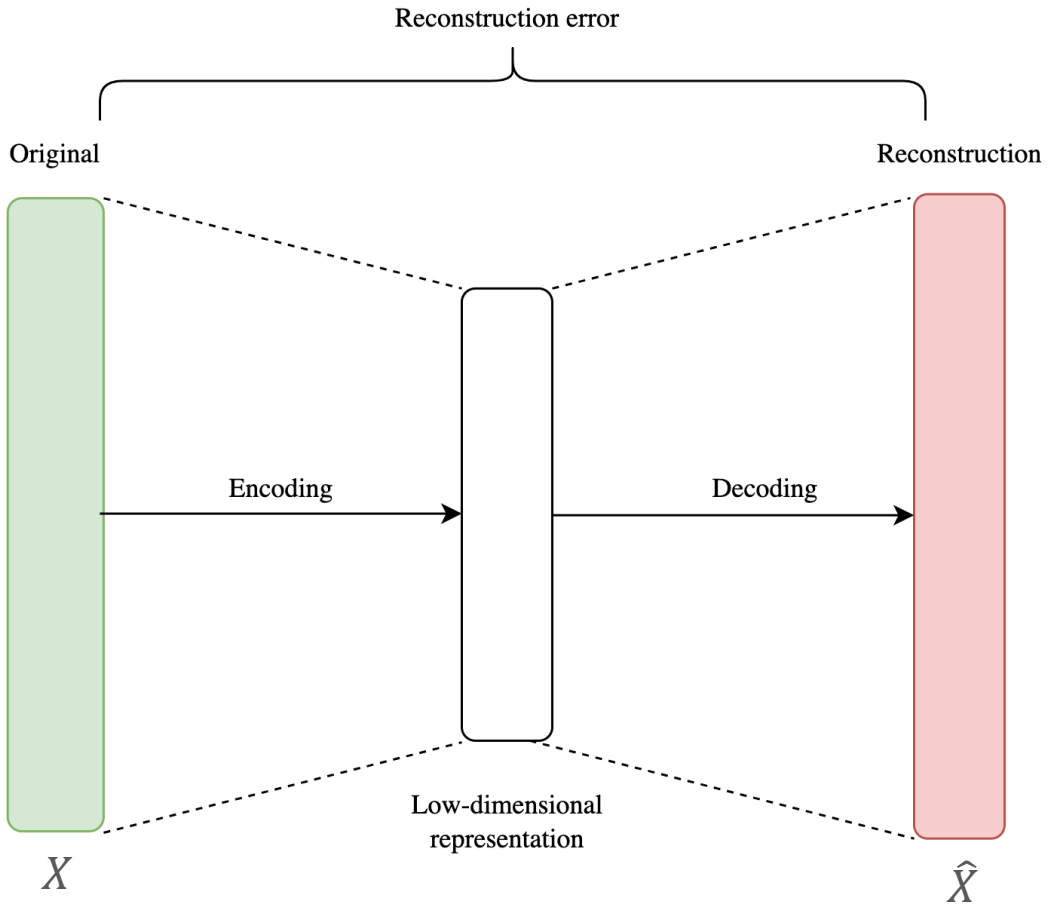
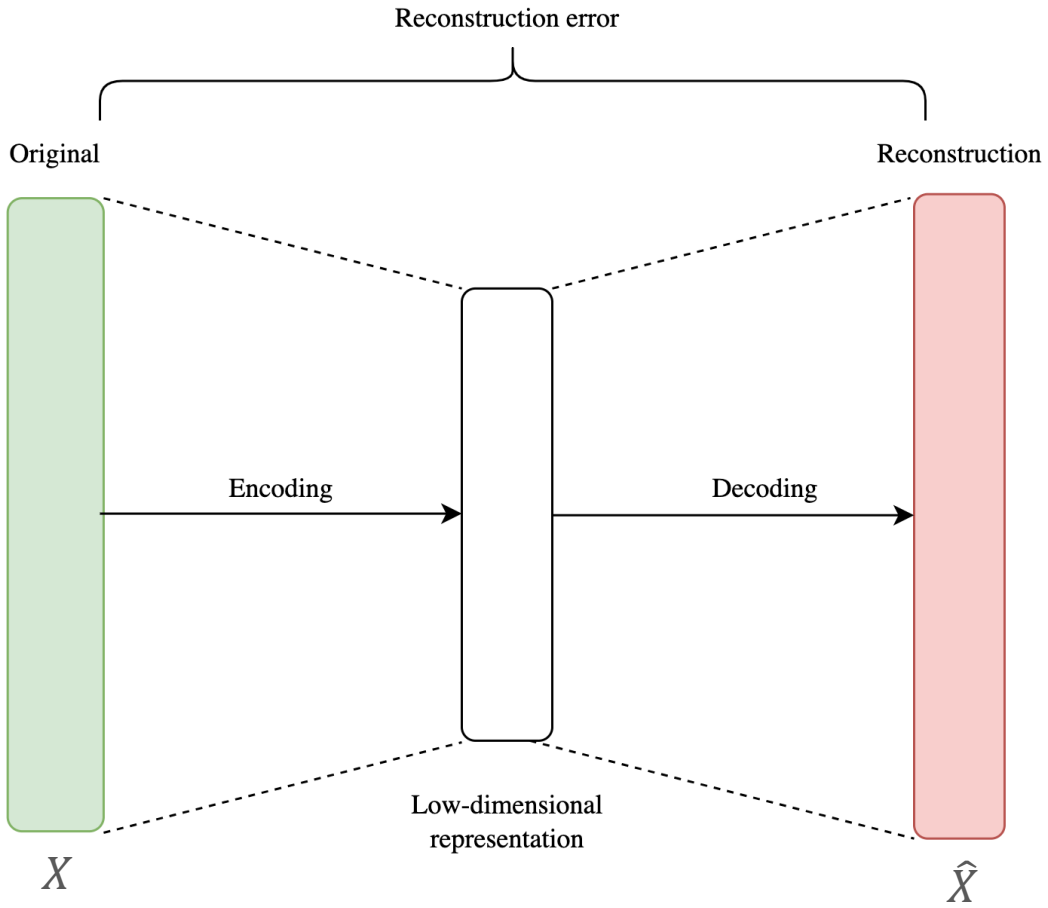


Image source: https://ymohamedahmed.github.io/post/pca/

AD with Principal Component Analysis (PCA)

X can be of any data modality

- Tabular data
- Univariate time series
- Multivariate time series
- Image



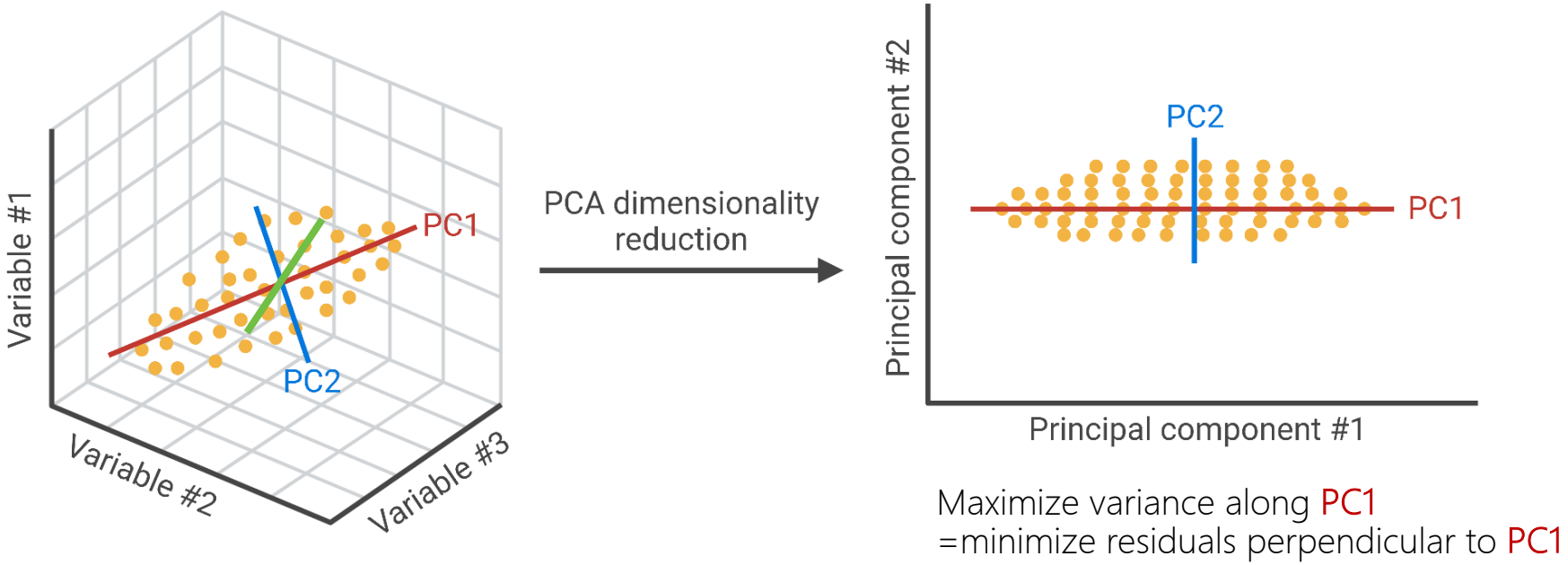
$$r_j = X_j - \hat{X}_j$$

Image source: https://ymohamedahmed.github.io/post/pca/

Reminder: dimension reduction with PCA

PCA can be viewed as fitting a $\mathbf{d}$-dimensional ellipsoid to the data:

- fit a d-dimensional ellipsoid to the data using a linear transformation.
- Large ellipsoid axes → high variance
- Small axis → variable does not explain much → less important → drop it.



Maximize variance along PC1
=minimize residuals perpendicular to PC1

Image source: https://www.biorender.com/template/principal-component-analysis-pca-transformation

Reminder: dimension reduction with PCA

- $X^{(0)}$ is the data matrix (n samples, d features).
- Center the data: $x_{ij} = x_{ij}^{(0)} - \mu_j$
- Calculate covariance matrix $\Sigma = \frac{1}{n-1} X^T X = \frac{1}{n-1} \begin{bmatrix} x_{11} & \cdots & x_{1d} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nd} \end{bmatrix}^T \begin{bmatrix} x_{11} & \cdots & x_{1d} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nd} \end{bmatrix}$
- Σ_{ij} is the cov. Between feature i and j.
- Σ_{ii} is the variance of feature i.
- Calculate the eigenvectors v_m and eigenvalues λ_m of Σ.

Independent directions along which the data varies the most

How much the data varies along this direction

- Sort the eigenvectors by decsending eigenvalues.
- Select the top $\mathbf{k}$ eigenvectors with the highest eigenvalues.
- Form the projection matrix $W = [v_1, v_2, \ldots, v_k]$
- The transformed data is $\mathrm{Z} = \mathrm{X}\, W$

$(n \times k)$ $(n \times d)$ $(d \times k)$

AD with Principal Component Analysis (PCA)

- Reduce the dimension of the data to a lower dimensional space.

$$Z = X W$$

- Reconstruct the data from the reduced space.

$$\hat{X} = Z W^T$$

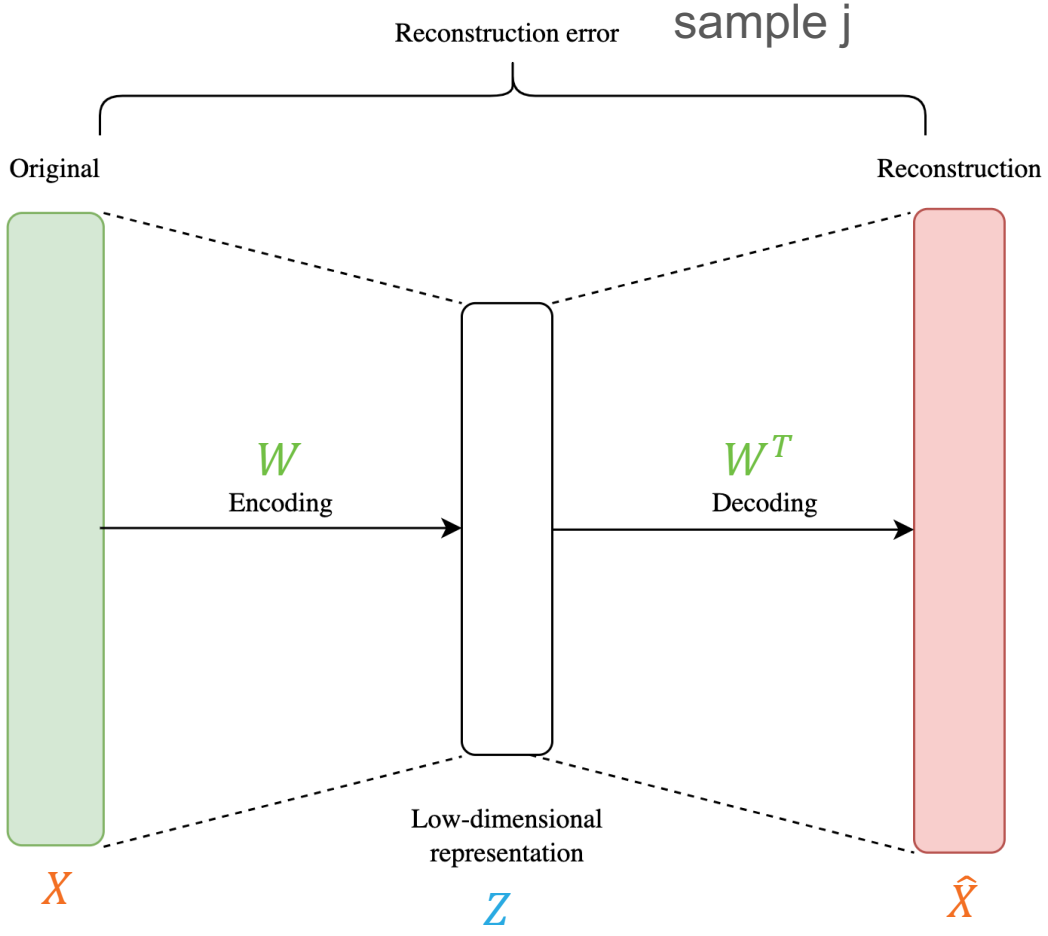
- Use only normal data to derive the PCA transformation.

➔Reconstructing abnormal samples will be unsuccessful, assuming they are generated by a different process.

➔Reconstruction errors can be used as anomaly scores.

$$R = X - \hat{X}$$

$r_j = x_j - \hat{x}_j$ → $\|r_j\|^2$ → anomaly score of sample j



$$\hat{X} = XWWT$$

Image source: https://ymohamedahmed.github.io/post/pca/

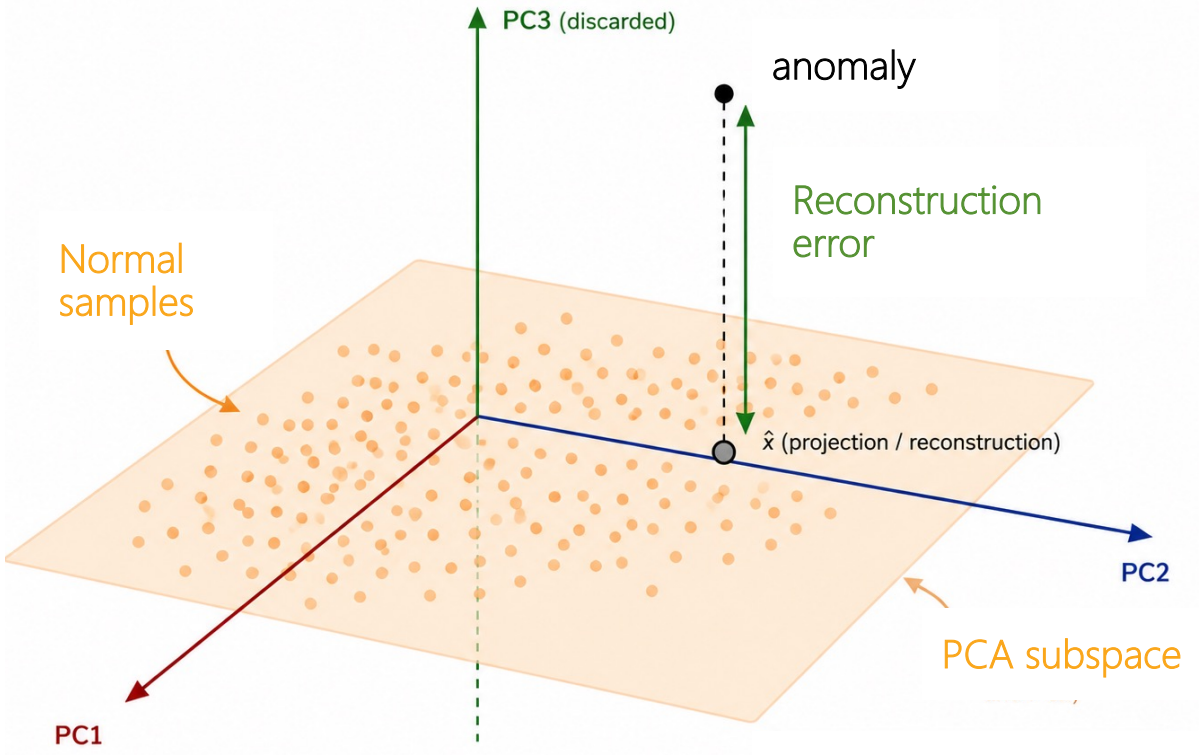
Why does reconstruction work?

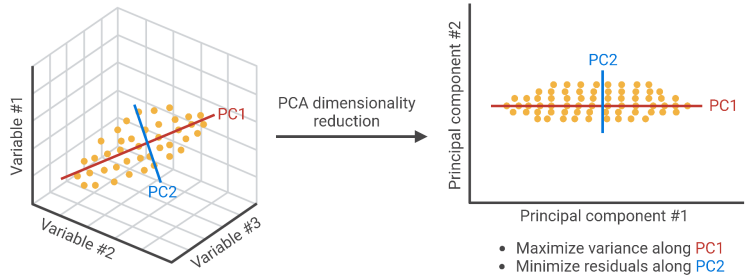
Intituition

- Normal data approximately lie in a low dimensional subspace.
- PCA estimates this subspace.
- Samples away from this subspace contain information that is lost at reconstruction.

→Their reconstruction is inaccurate.

The reconstruction error is the distance to the projected subspace of the normal data.





How to select the number of PCs?

- Keeping all dimensions → prefect reconstruction of both normal and anomalies
- Strong compression → no reconstruction possible even for normal data

→Some compression is needed, not too much.

→ No. of PCs is a hyperparameter
→ Tune on validation data if available
→ Use cross validation, then fix before testing.

Note: the explained variance is indicative of the normal reconstruction ability, not necessarily of detection quality.

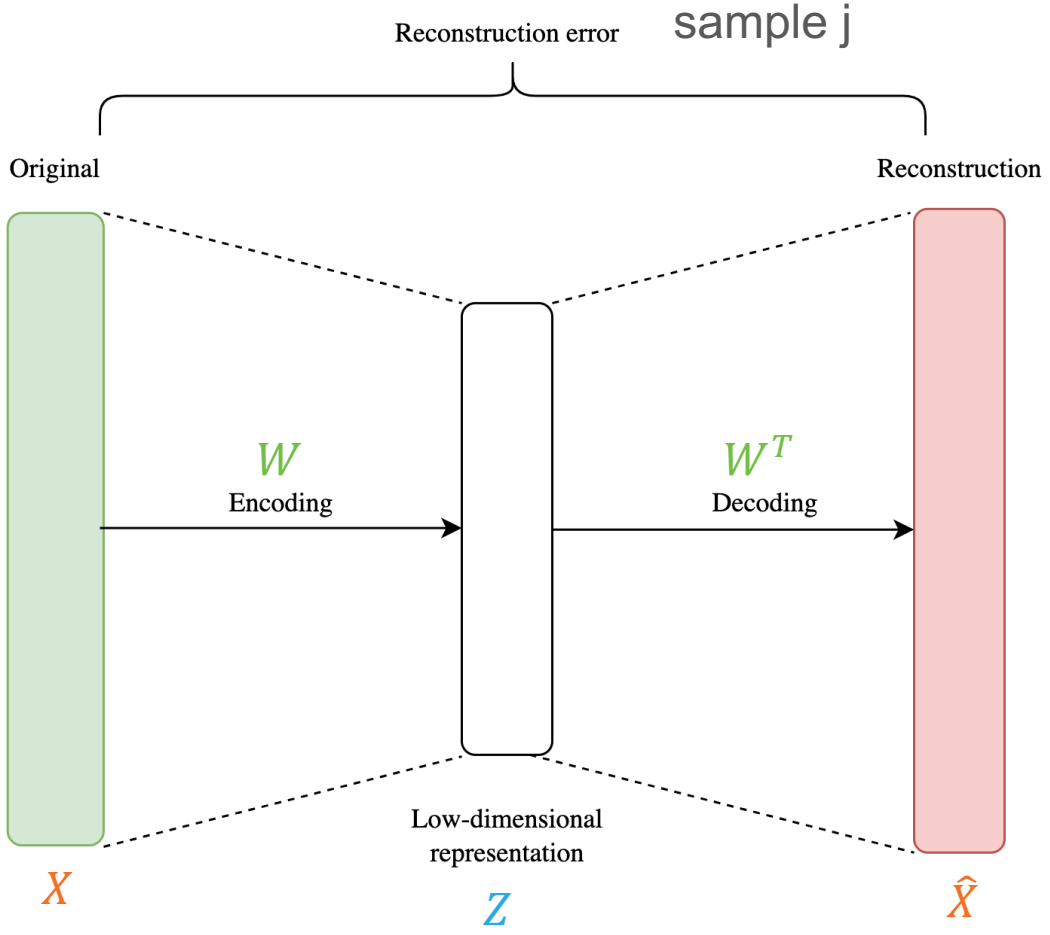
PCA in practice

- Training
  - Fit PCA on healthy data only
  - Standardize features (why?)
- Anomaly score can be designed:
  - Reconstruction MSE
  - Max residual
  - …
- Threshold
  - Use validation data if available
  - Percentile
  - Robust statistics
- Advantages
  - Very fast training and inference
  - One hyperparameter
  - Interpretable

Can inspect

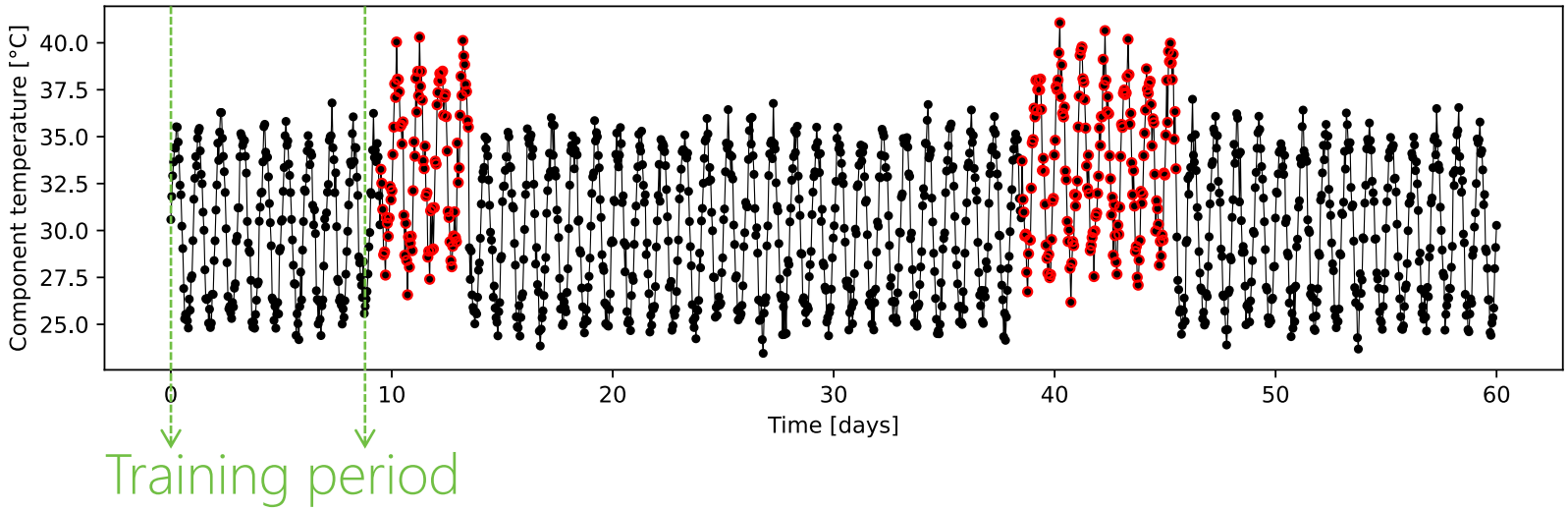
- The contribution of each variable to each PC→ W
- The reconstruction residuals of the individual variables (partially interpretable, as errors can propagate between correlated variables)
- Understand the complexity of the normal data (sensitivity to the number of PC).

$r_j = \mathrm{x}_j - \hat{\mathrm{x}}_j$ → $\|r_j\|^2$ → anomaly score of sample j

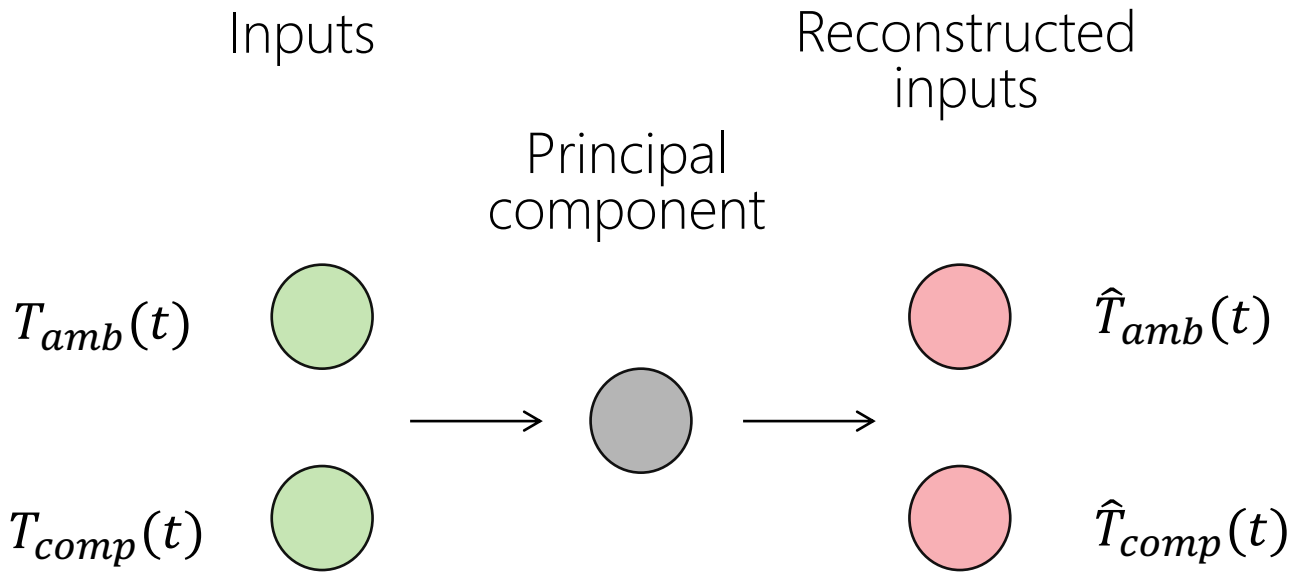


Example: anomalous periods of component temperature

- Assume a first normal period.
- Train PCA on it.
- Apply the transformation on all data.
- Calculate reconstruction errors.

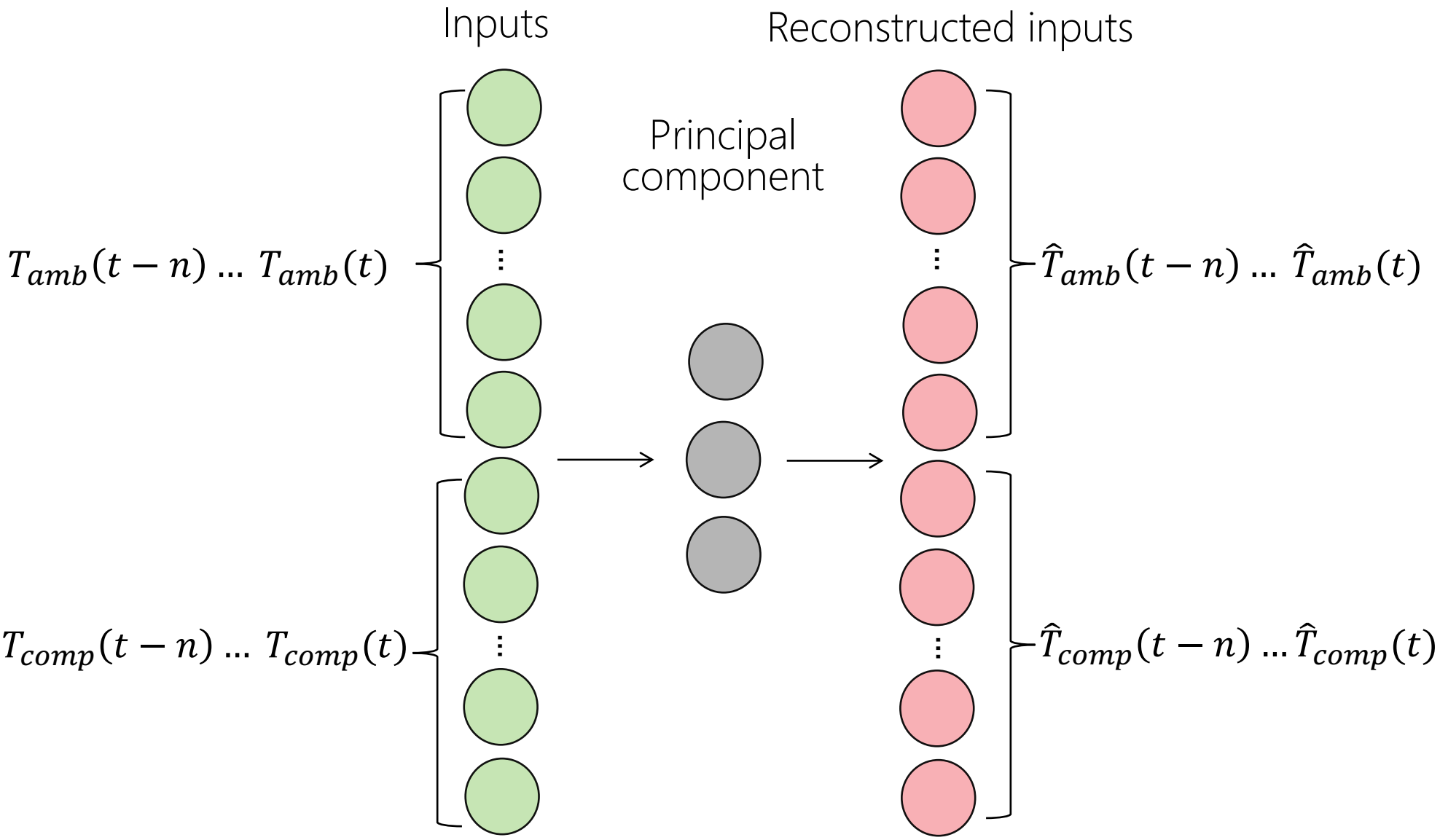


PCA with single observations (tabular data)

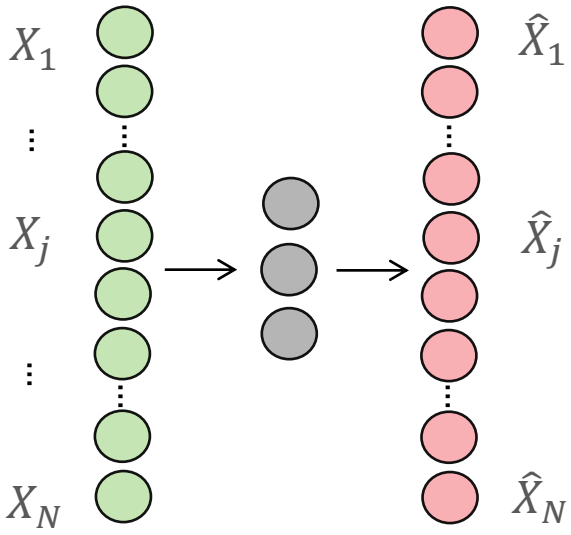


Question: why is this equivalent to tabular data (e.g like individual transactions of a credit card)?

PCA with (multivariate) time series

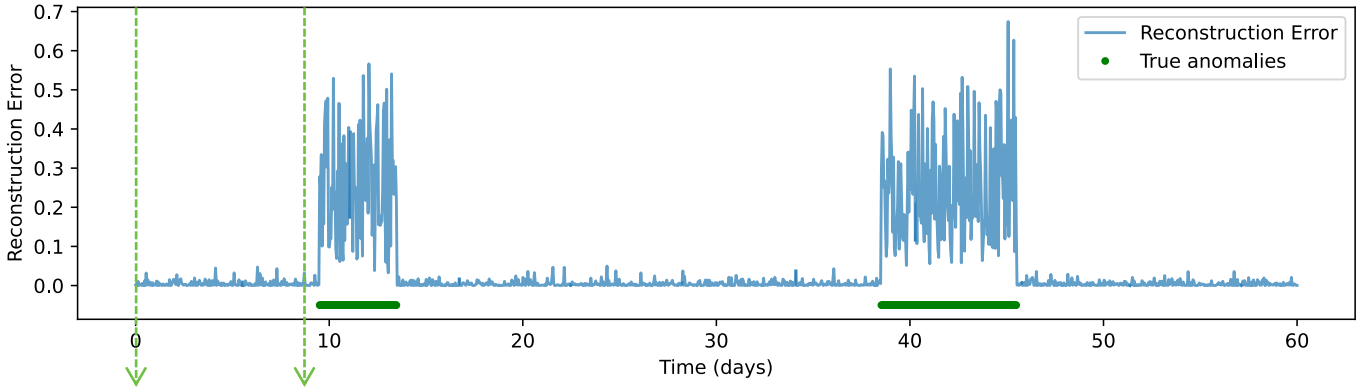


Reconstruction errors

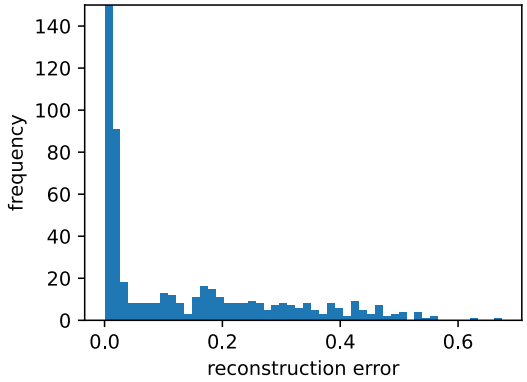


$$r_j = X_j - \hat{X}_j$$

$$Reconstruction\ Error = MSE = \frac{1}{N}\sum_{j=1}^{N}\|X_j - \hat{X}_j\|^2$$



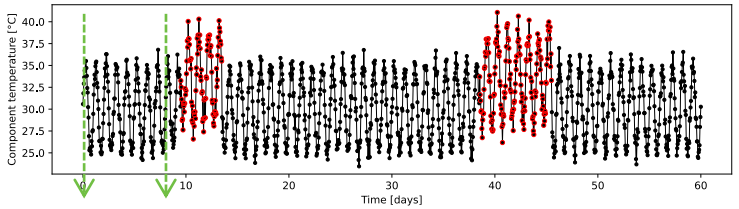
Training period



WPI_AnoDet_05a_PCA_time_series

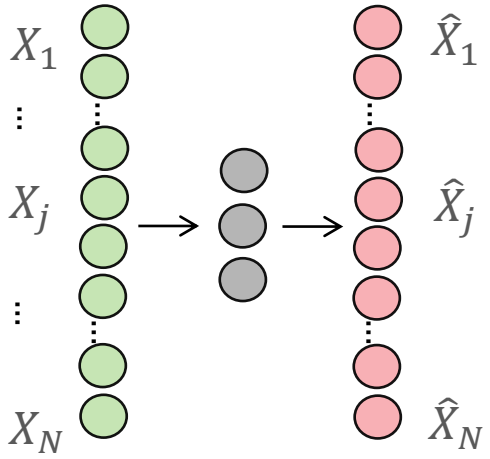
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  - Use validation data if available
  - Percentile
  - Robust statistics
- Advantages
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Training

$$MSE = \frac{1}{N}\sum_{j=1}^{N}\|X_j - \hat{X}_j\|^2$$



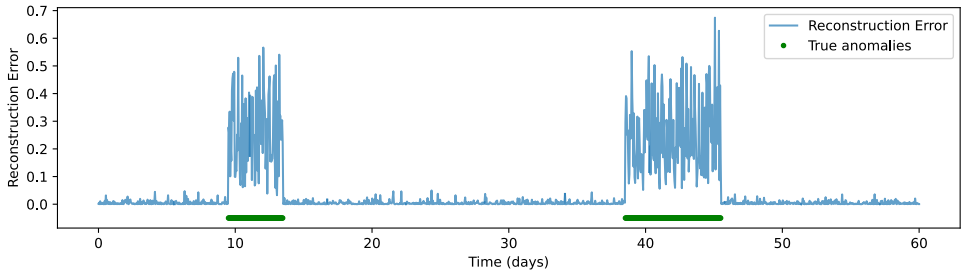
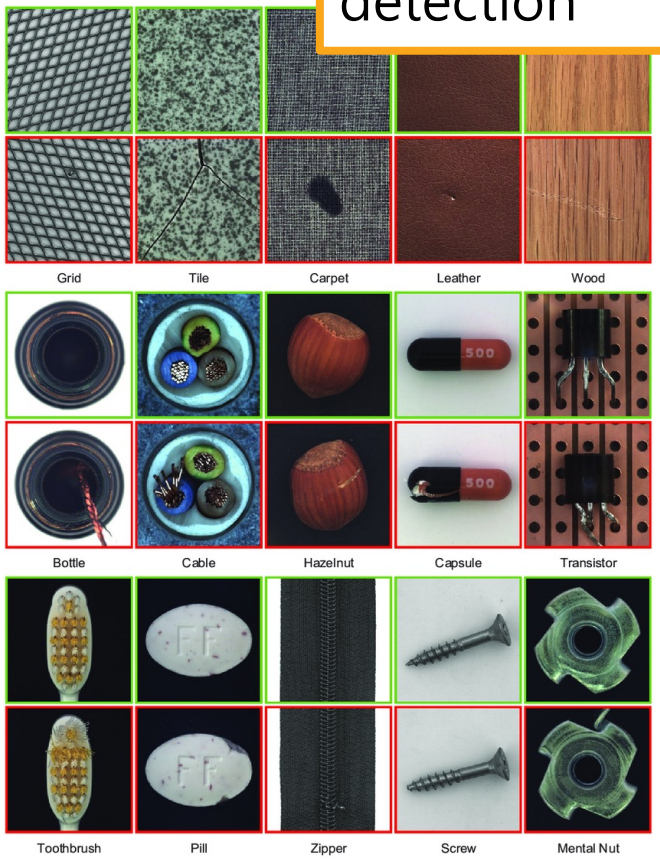


Image Anomaly Detection

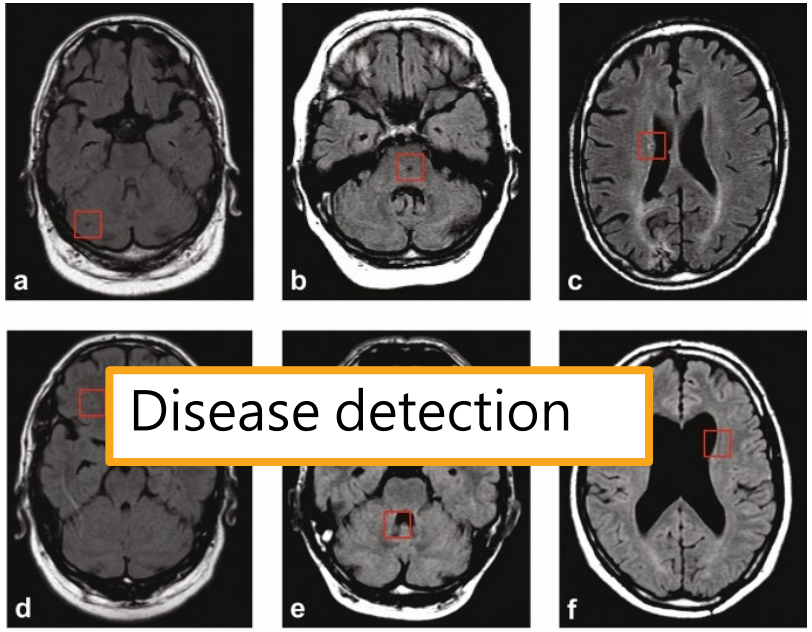
Damage/Defect detection



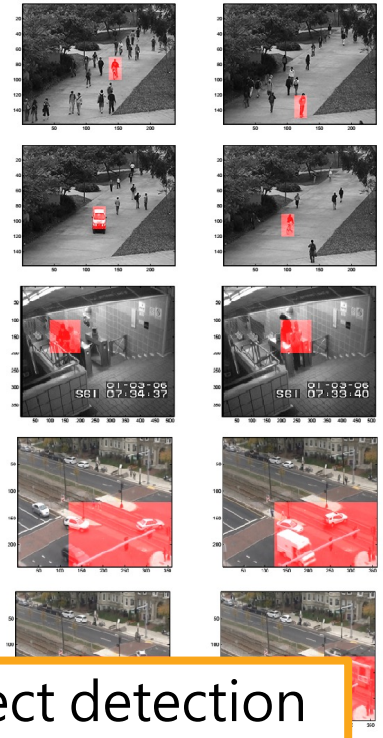
Production quality



Disease detection

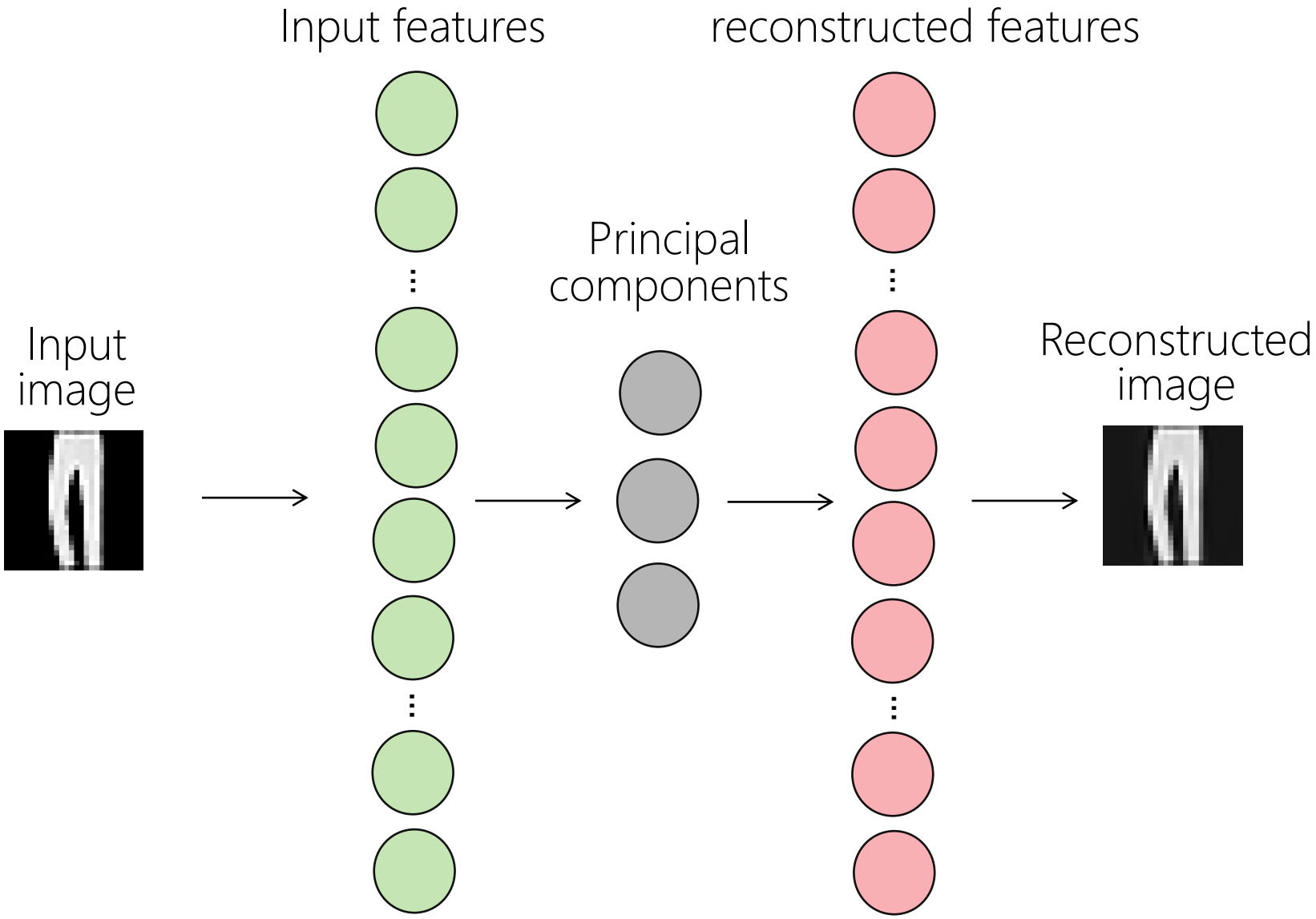


Object detection



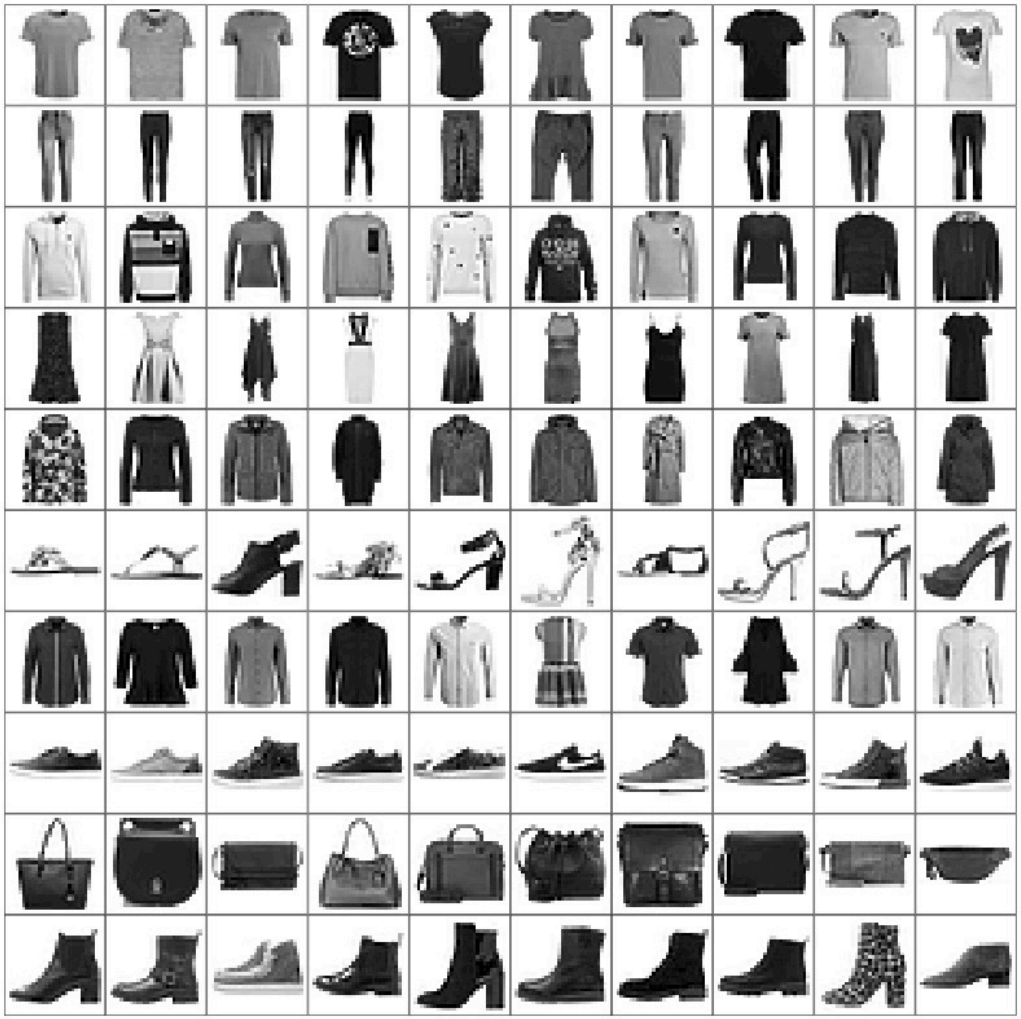
Note: we focus on detection, not segmentation.

Image Anomaly Detection with PCA



Fashion-MNIST Dataset

- Created by Zalando
- 10 article classes
- 60000 training images
- 10000 test images
- Typically used for classification benchmarking.
- Can be used for image anomaly detection benchmarking.



Adapt F-MNIST for AD

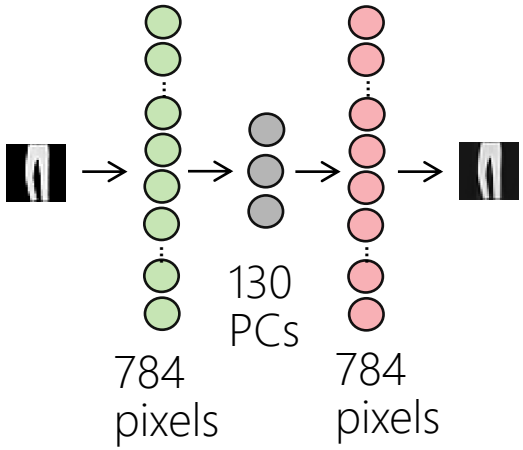
- One vs. One:
  - <trousers > is the majority = normal.
  - <T-shirt> is the minority = abnormal.
- One vs. All:
  - <trousers > is the majority = normal.
  - A mix of all other articles is the minority = abnormal.

- "Clean" training: use images of the normal class to train a PCA.
- At inference: other articles will be badly reconstructed

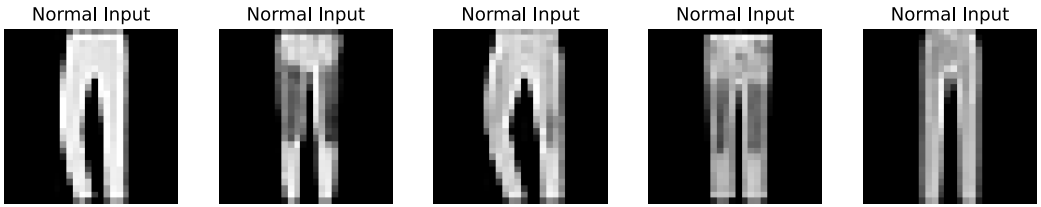
➔ Recon. Errors can serve as anomaly scores.

Image Anomaly Detection with PCA

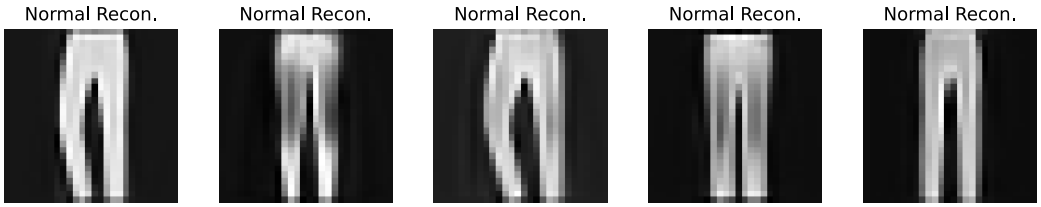
Example: trousers vs. all



Normal images

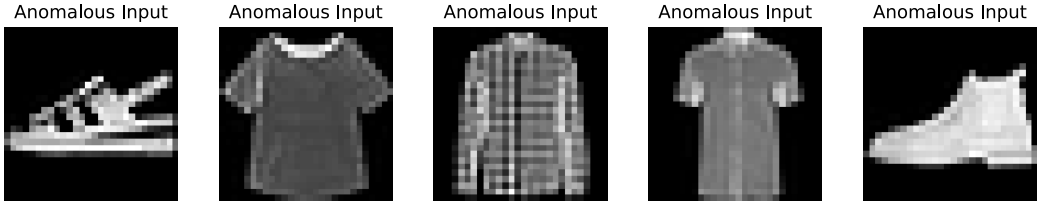


input

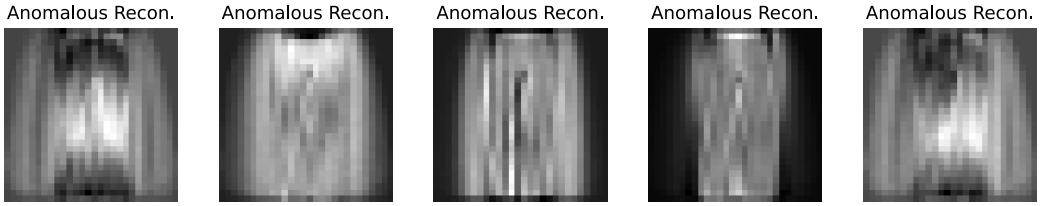


reconstruction

Abnormal images



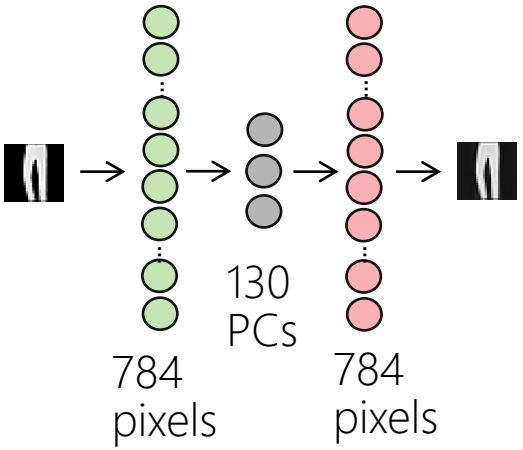
input



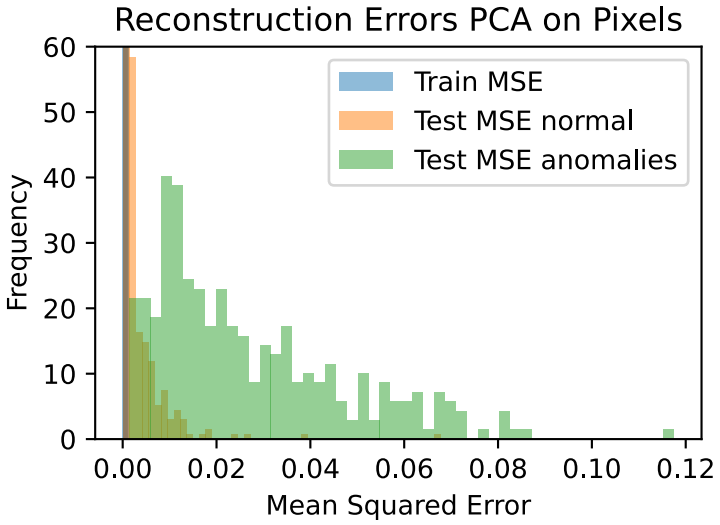
reconstruction

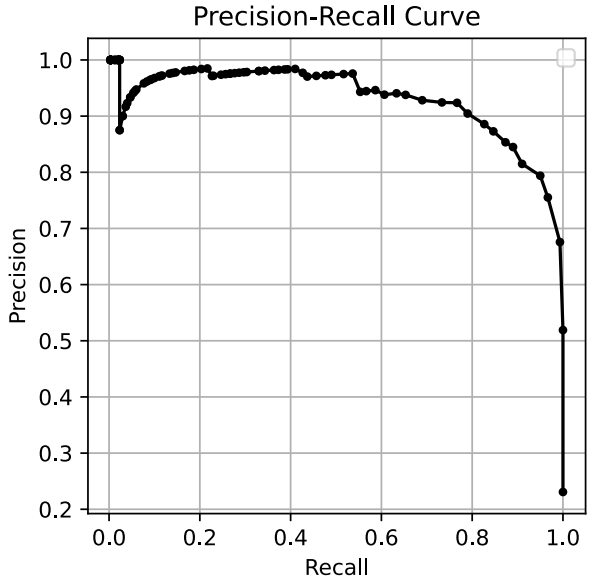
Image Anomaly Detection with PCA

- Example: trousers vs. all
- 130 principal components.
- Calculate the mean squared reconstruction error (MSE) for each image.



$$MSE = \frac{1}{N}\sum_{i=1}^{N=784} \left\|X_i - \hat{X}_i\right\|^2$$





WPI_AnoDet_05b_PCA_FMNIST.

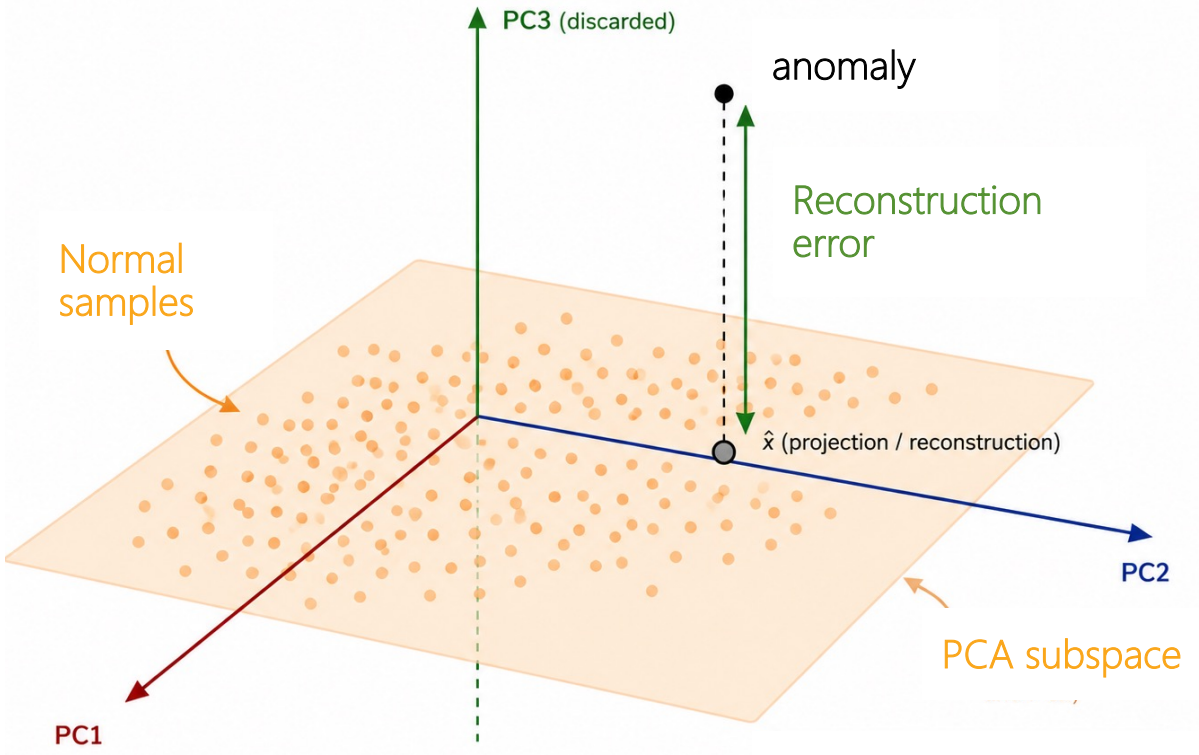
Why does reconstruction work?

Intituition

- Normal data approximately lie in a low dimensional subspace.
- PCA estimates this subspace.
- Samples away from this subspace contain information that is lost at reconstruction.

→Their reconstruction is inaccurate.

The reconstruction error is the distance to the projected subspace of the normal data.



Can anomalies have a small reconstruction errors?

Yes, if

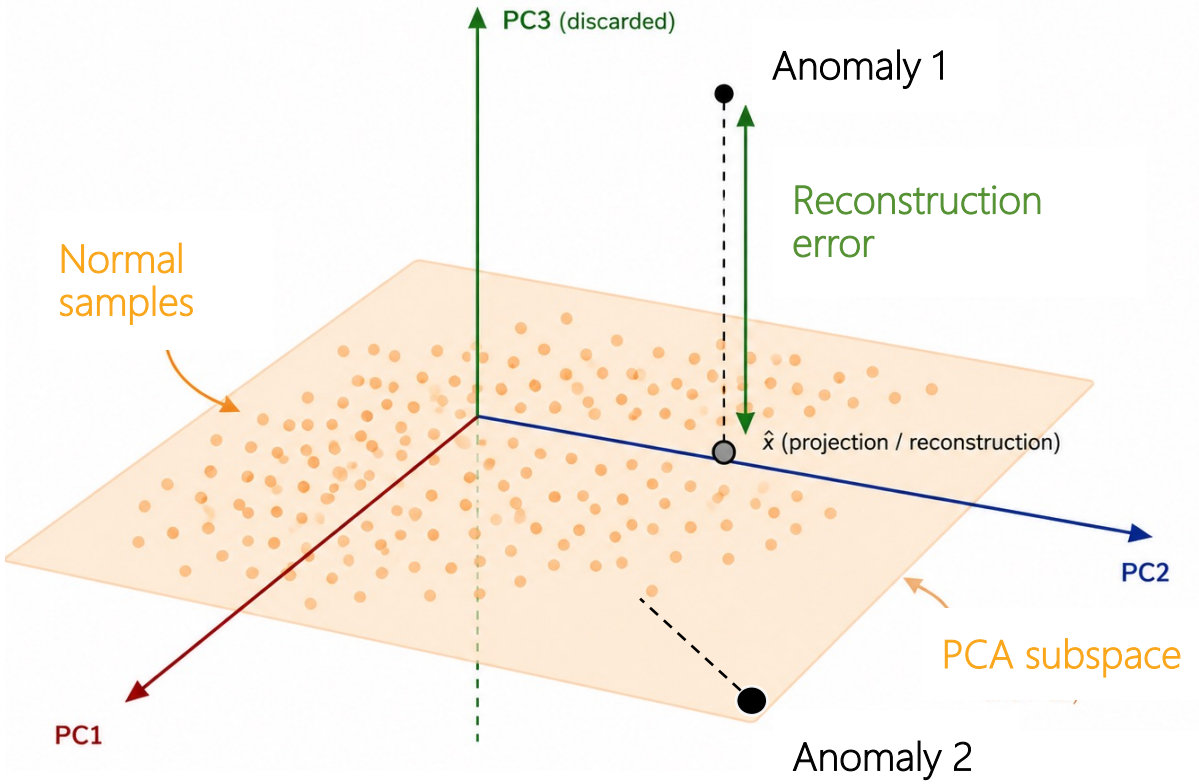
1. The dimension compression is too weak
2. The anomaly is mostly within the reduced subspace, and has a large distance from the cloud inside the plane.

→2 types of anomalies:

1. Point above the plane → large residual → detected
2. Point far away inside the plane → small residual → not detected.

Measure distance within the plane

→Hotelling's T^2 statistics and use as an additional threshold



Limitations of PCA

- Requires normal training data – sensitive to contamination
- Reconstruction errors may propagate between related variables
- Linear model: performance degrades when linear transofrmation of the normal data is not enough.

→ Reconstruction neural networks = autoencoders

Autoencoder neural networks

An autoencoder is a neural network trained to reconstruct its input.

Wind speed

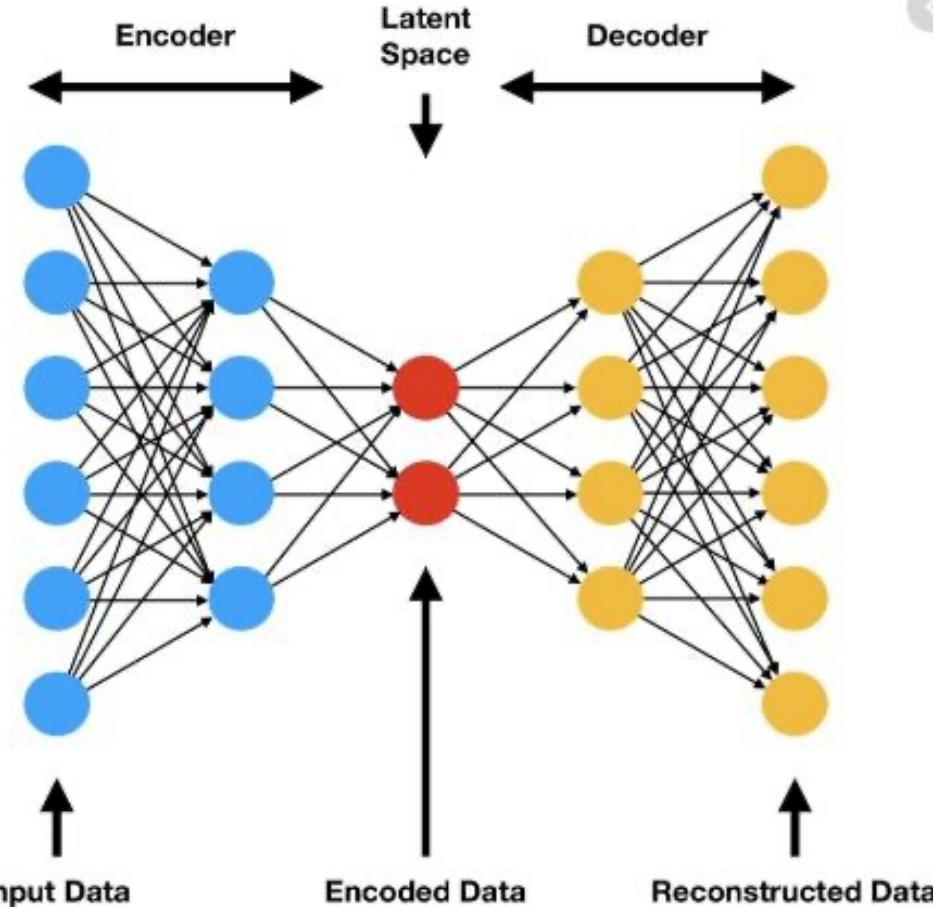
Ambient temp.

Rotor speed

power

Gen. Bear Temp.

Gear. Oil Temp.



Residual = reconstruction error:

$$r_j = X_j - \hat{X}_j$$

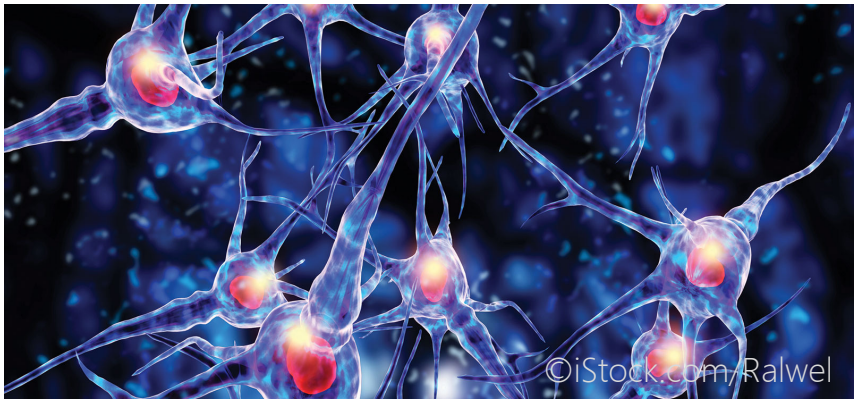
↓

Anomaly score

Z Yang, et al (2021)."Autoencoder-based representation learning and its application in intelligent fault diagnosis: A review

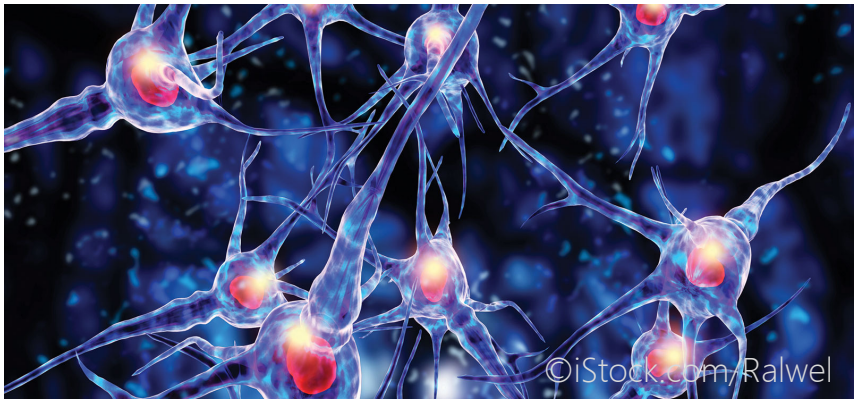
What is a neural network?

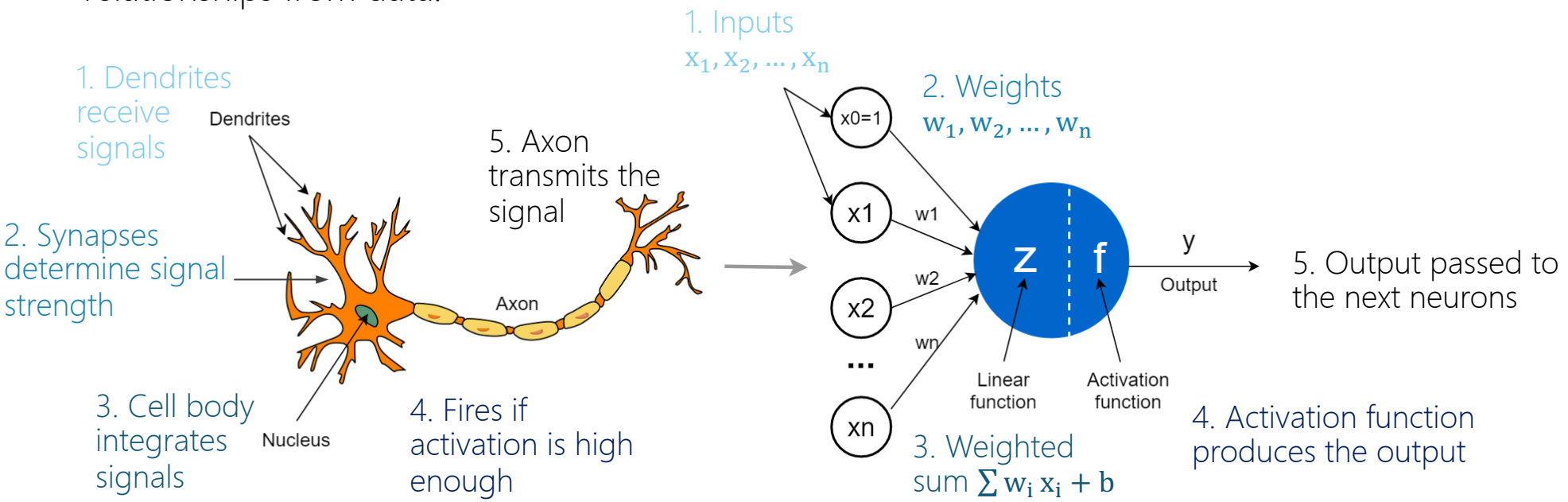
- A neural network is a machine learning model inspired by the way biological neurons exchange signals.
  - It consists of many simple computational units ("neurons") connected together.
  - Each neuron performs only a simple calculation.
  - By combining thousands or millions of neurons, the network can learn highly complex relationships from data.



What is a neural network?

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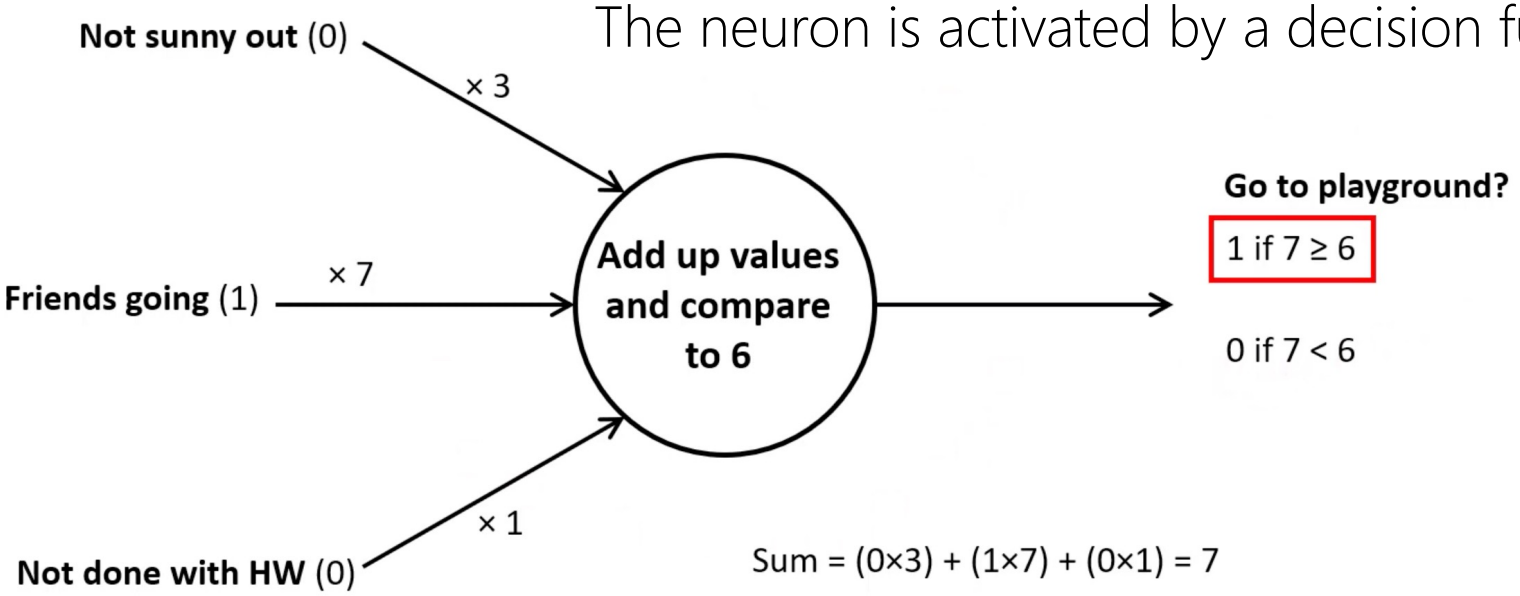




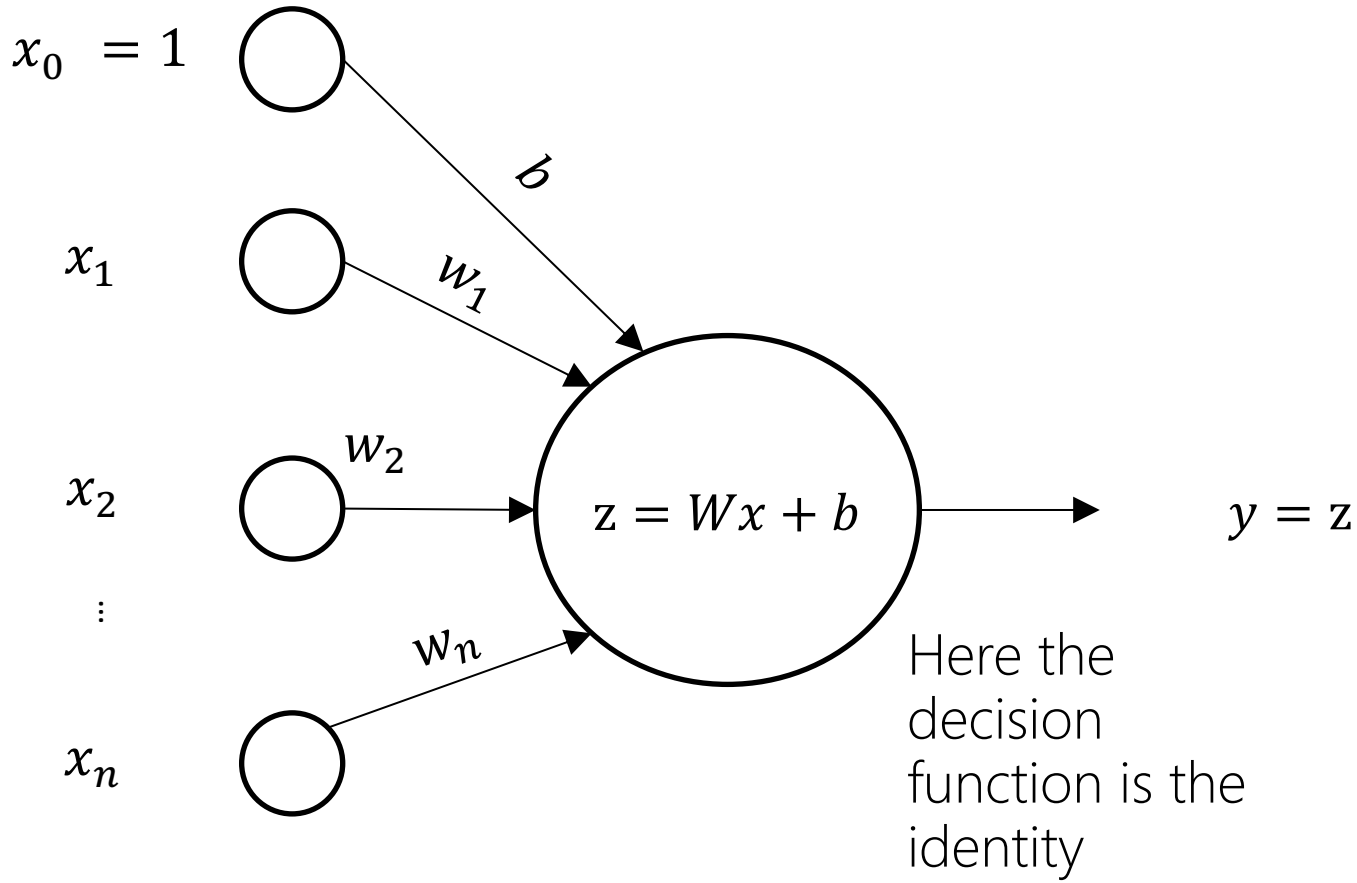
https://towardsdatascience.com/the-concept-of-artificial-neurons-perceptrons-in-neural-networks-fab22249cbfc/

- Neural networks are computer systems inspired by the human brain.
- They learn to recognize patterns and make decisions.
- They are composed of layers of artificial neurons that are interconnected.
- Learning is achieved by adjusting the connections based on data.

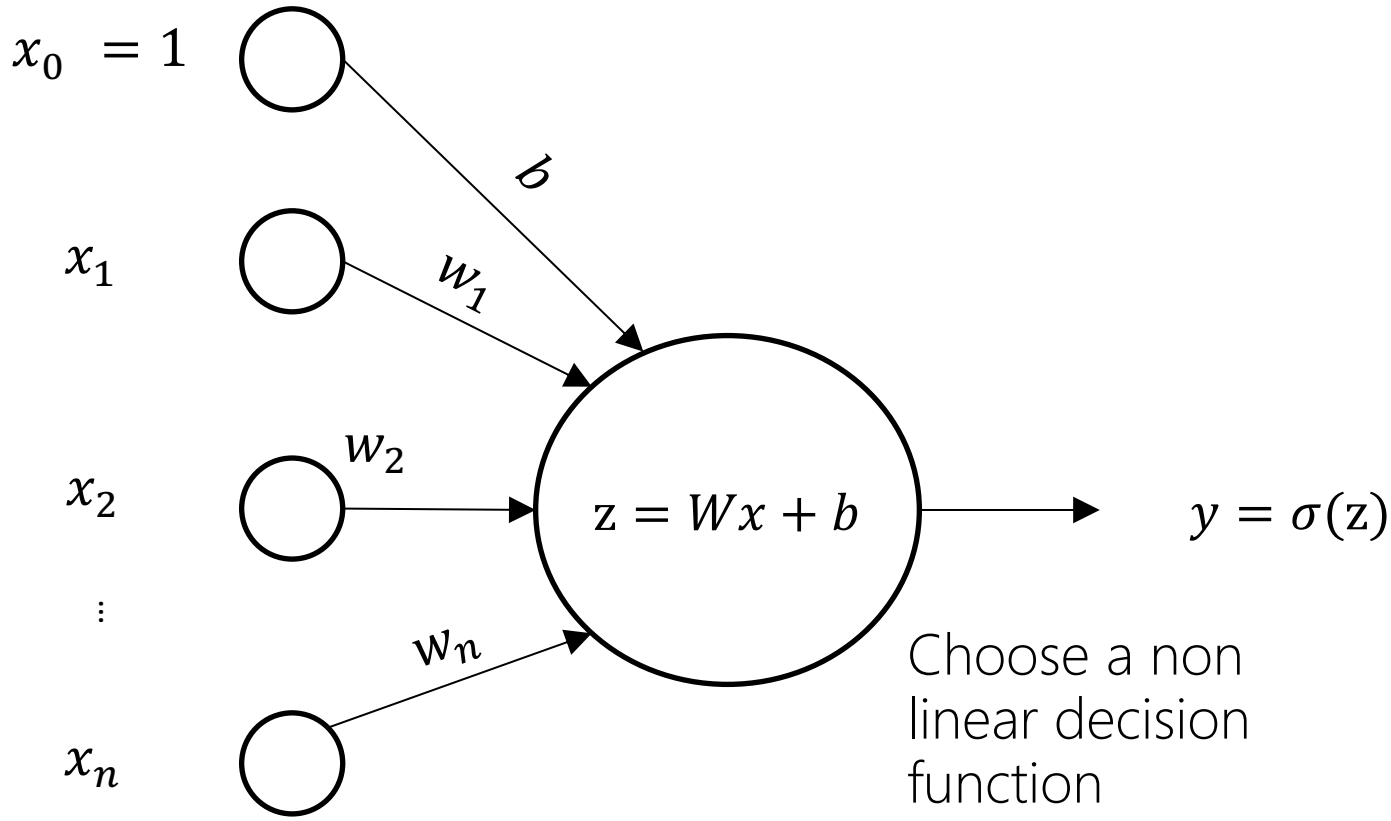
Imagine a neuron as a decision unit
The neuron is activated by a decision function

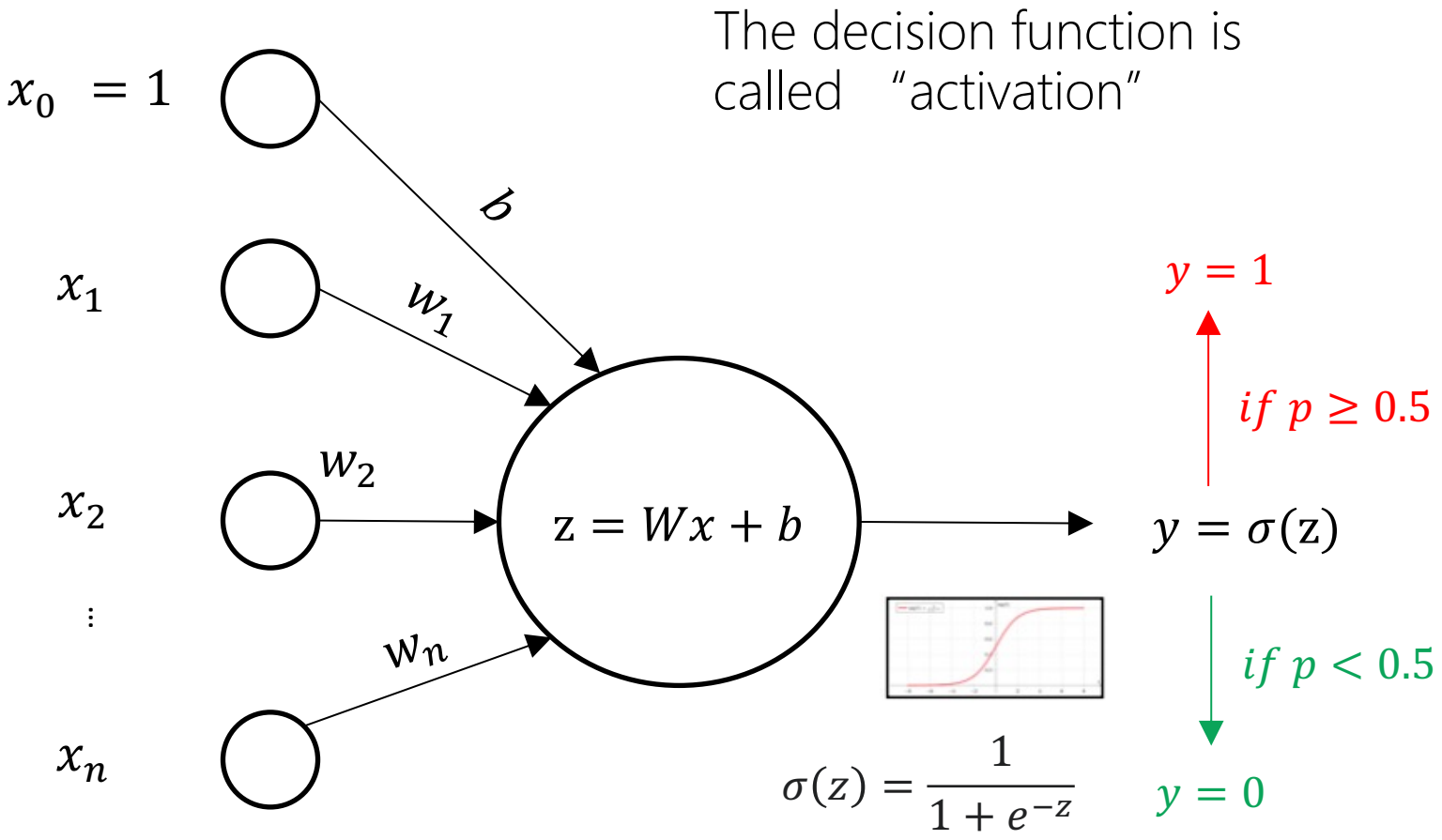


A Linear Regression as a NN



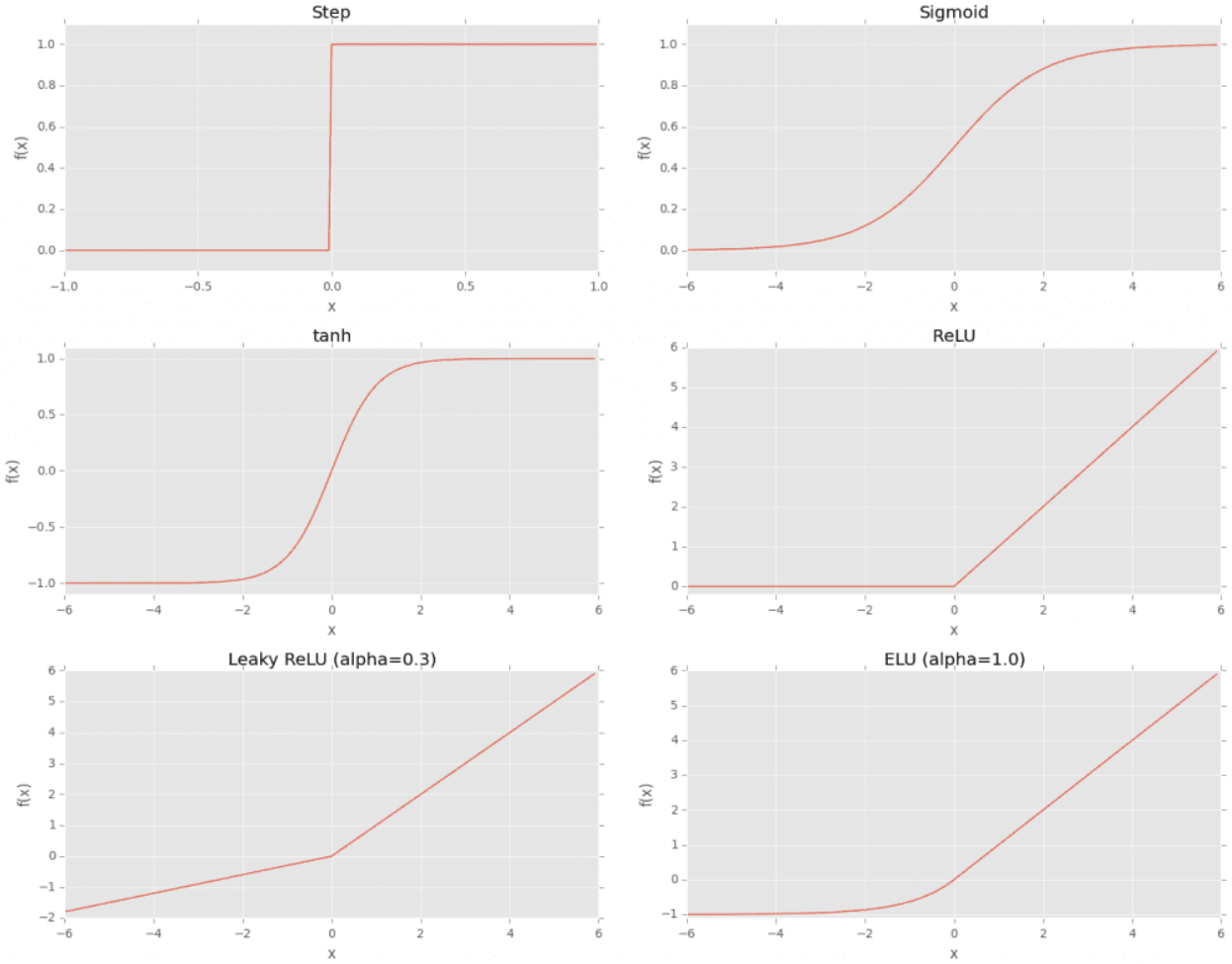
Non Linear Regression as a NN

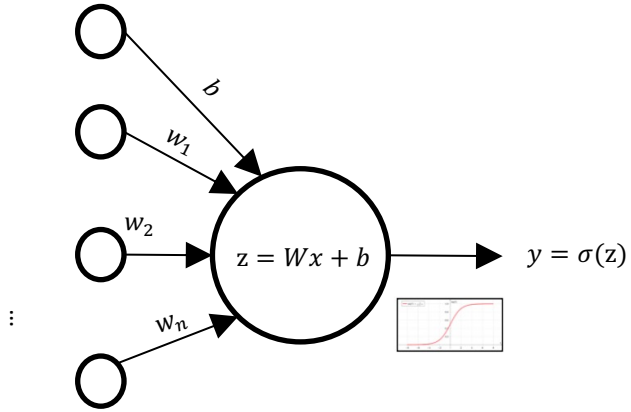




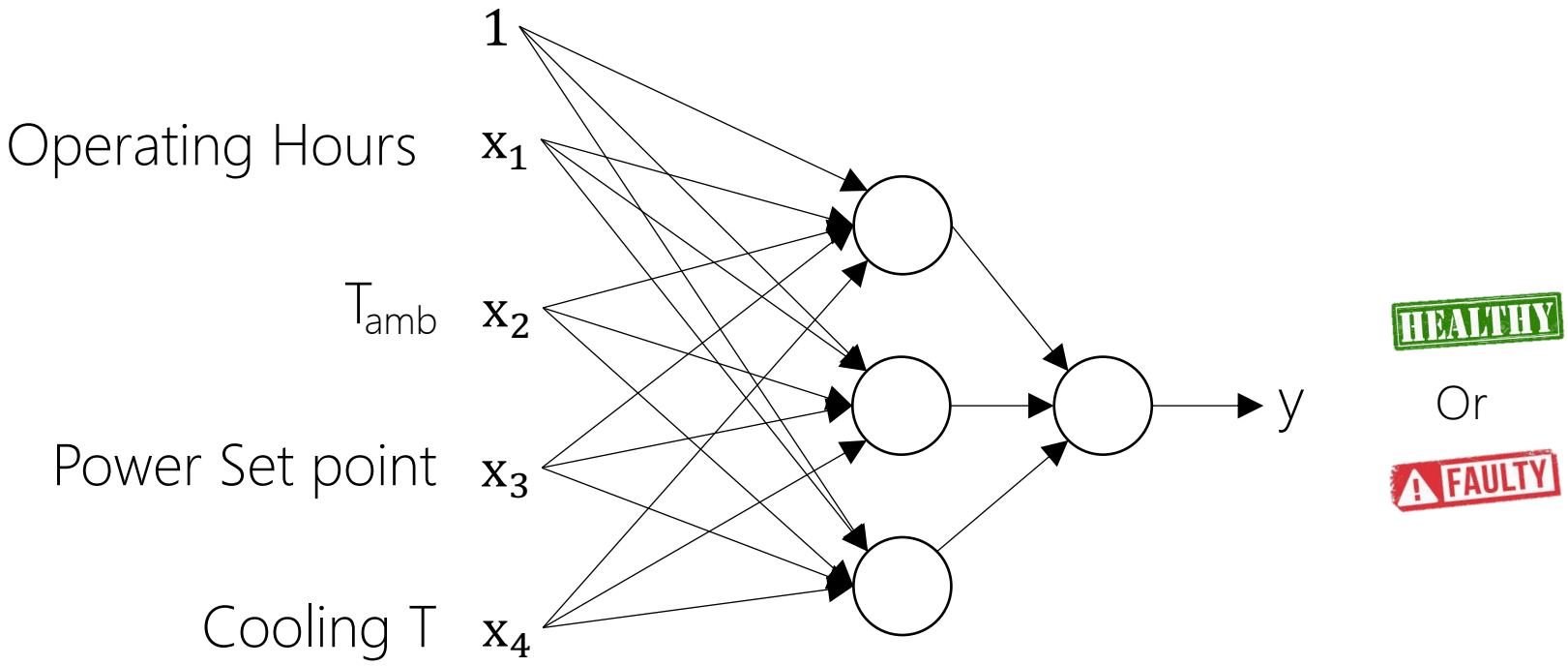
A NN for a binary outcome with only 1 neuron with sigmoid-activation is nothing else than logistic regression!

Activation Functions



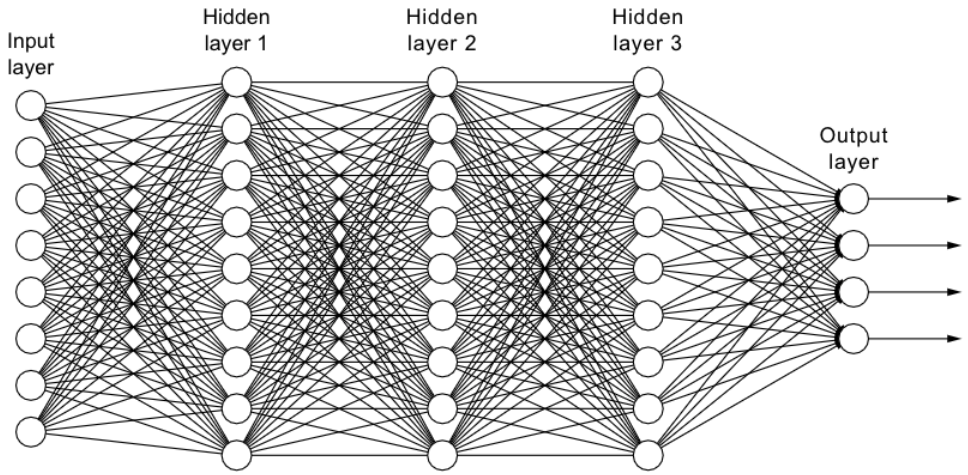


NNs with Hidden Layers

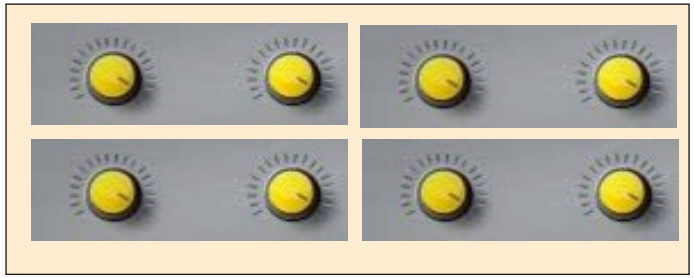


To model complex relationships we need complex functions ➔ achieved with hidden layers + non linear activations

Learning Process: How to Determine Weights?



Given the input, the output of a NN is defined by the weights



Typical >1 Mio. weights

- During training the weights are tuned so that the NN best fulfils its specific task (classification, regression, reconstruction…).
- Tuning is achieved by minimizing a **loss function**.

- Which loss function we minimize? depends on the task.
- Main tasks of NNs predict y given x:
  - Classification
    - Predict a class label (e.g binary classification $Y \in \{0,1\}$)

    ➔Loss function: Cross Entropy
  - Regression
    - Predict a number Y.

    ➔ Loss Function Mean Squared Error (MSE).
  - Reconstruction
    - Reconstruct the input X

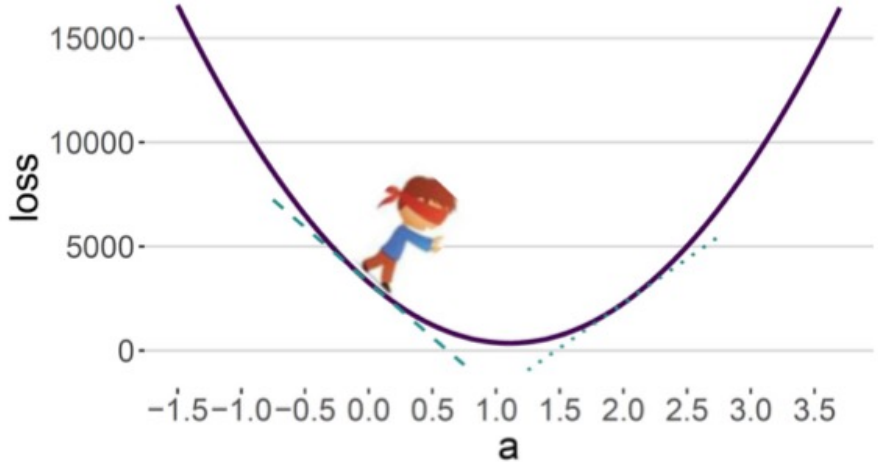
    ➔ Loss function MSE.

How is the Loss Minimized?

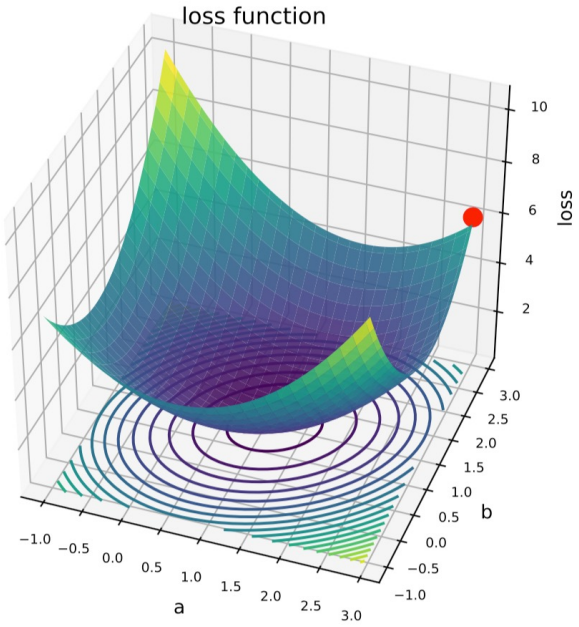
- Method: Gradient Descent.
- With 1 weight:
  - Take a large step if slope is steep (you are away from minimum)
  - Slope of loss function is given by gradient of the loss w.r.t. a $\frac{\partial L(a)}{\partial a}$
  - Iterative update of the parameter a

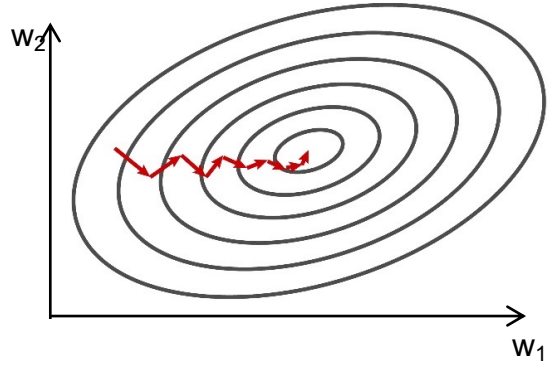
$$a^{(t)} = a^{(t-1)} - \varepsilon^{(t)} \frac{\partial L(a)}{\partial a}\Big|_{a=a^{(t-1)}}$$

learning rate



With 2 weights:



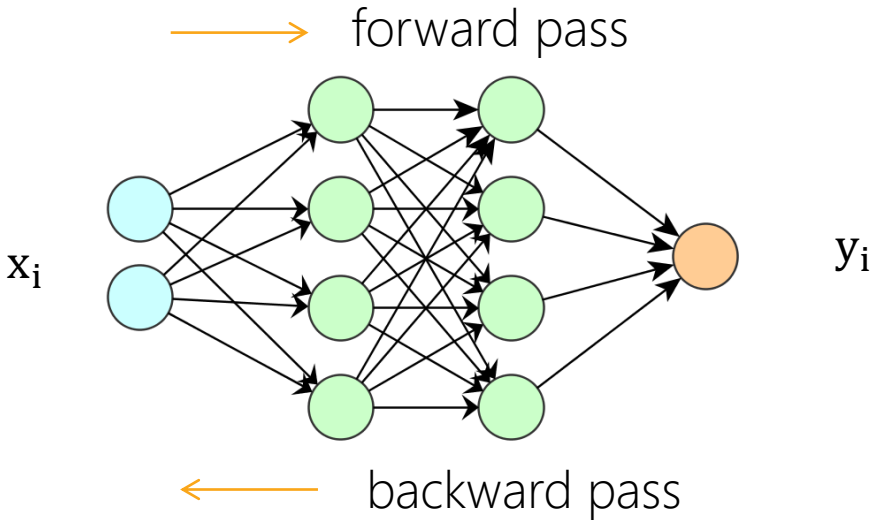


$$w_i^{(t)} = w_i^{(t-1)} - \varepsilon^{(t)} \left. \frac{\partial L(\mathbf{w})}{\partial w_i} \right|_{w_i = w_i^{(t-1)}}$$

We efficiently tune the weights in a NN via forward and backward pass

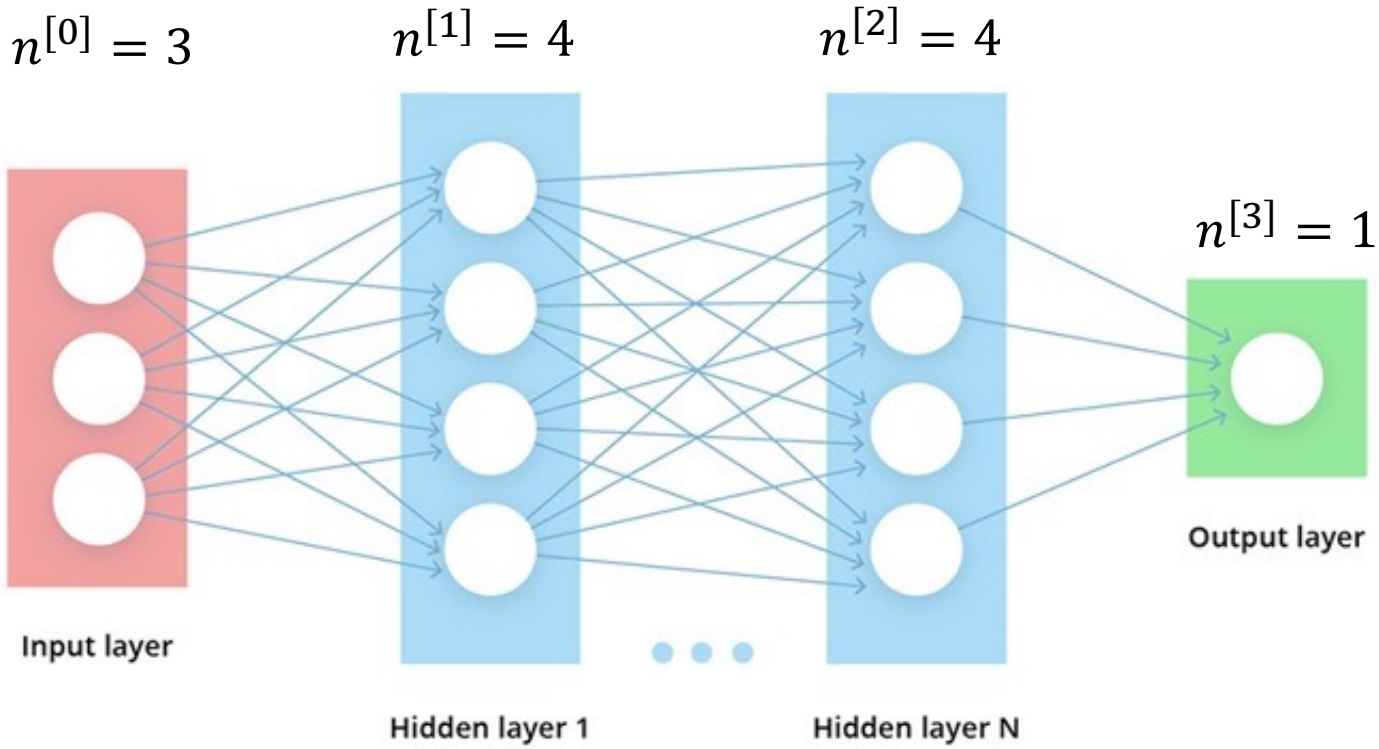
Forward pass propagates training examples through network

Predicts an output $\mathbf{y_i}$ for each input $\mathbf{x_i}$ in the batch given the NN weights $\mathbf{w}$
→ With the predicted and observed $\mathbf{y_i}$ we compute the loss L



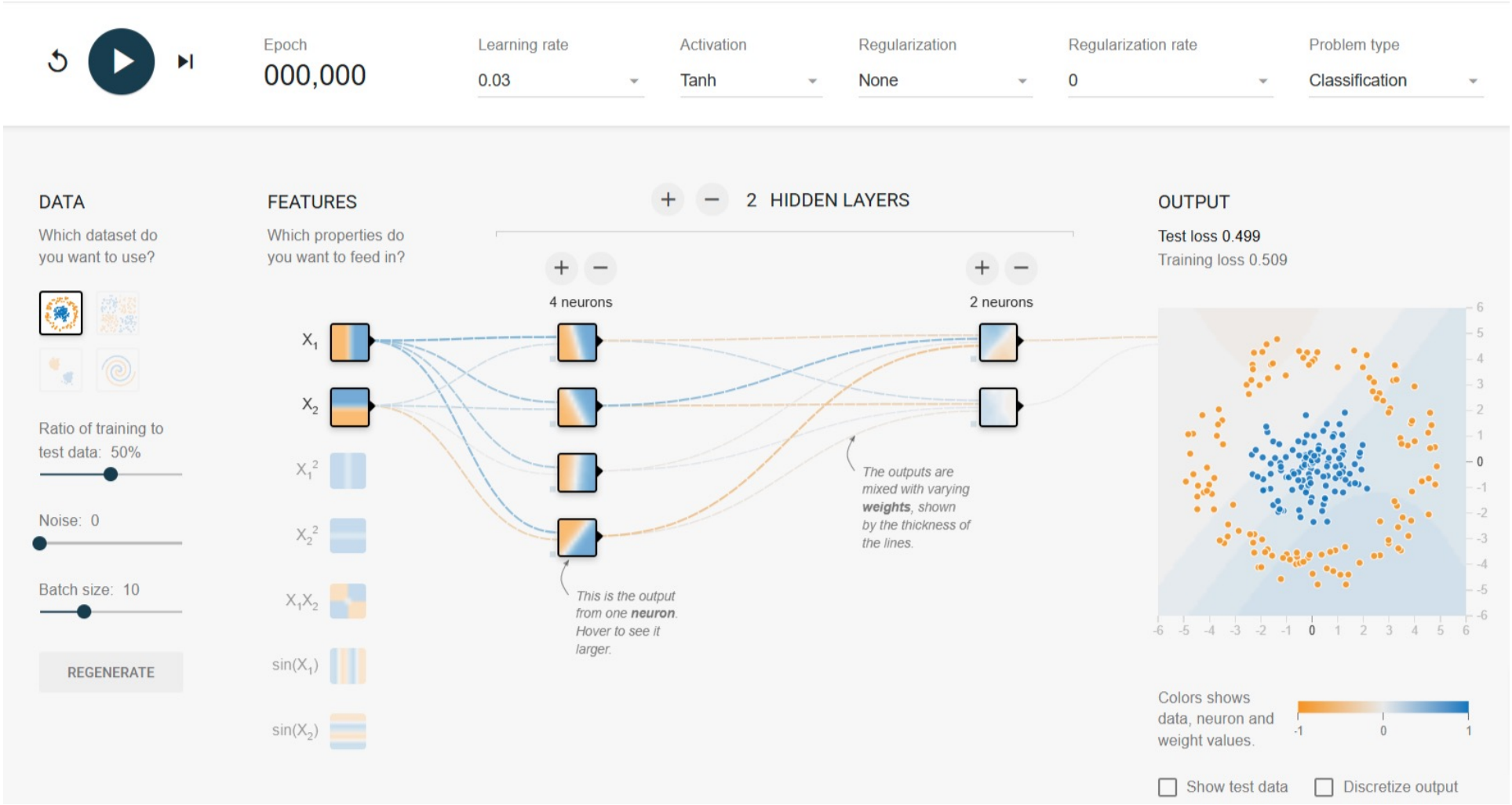
Backward pass propagates gradients through network

Via chain rule all gradients $\frac{\partial \mathbf{L(w)}}{\partial w_k}$ are determined → update weights $w_i^{(t)} = w_i^{(t-1)} - \varepsilon^{(t)} \frac{\partial L(\mathbf{w})}{\partial w_i}\Big|_{w_i = w_i^{(t-1)}}$



$x \in R^n$ $z^{[1]} = \sigma(w^{[1]}x + b^{[1]})$ $z^{[2]} = \sigma(w^{[2]}z^{[1]} + b^{[2]})$ $y = \sigma(w^{[3]}z^{[2]} + b^{[1]})$

Intro to neural networks – Neural Network Playground



[Neural Network Playground]

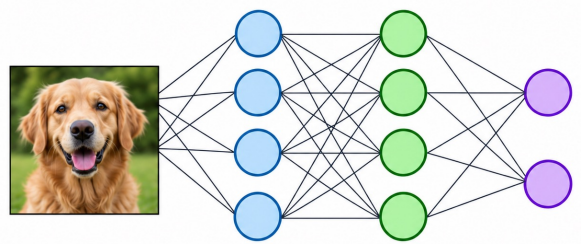
- Which loss function we minimize? depends on the task.

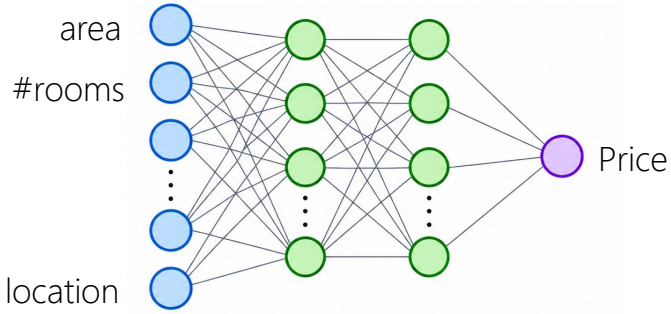
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  - Classification
    - Predict a class label (e.g binary classification $\mathbf{Y} \in \{0,1\}$)

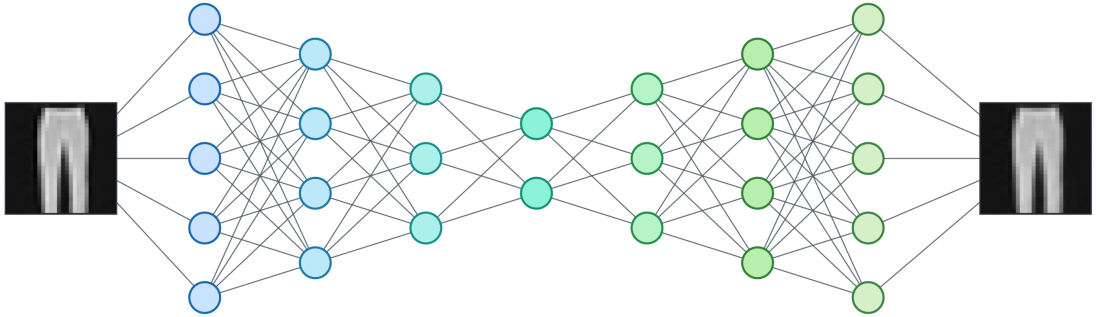
    ➔Loss function: Cross Entropy
  - Regression
    - Predict a number Y.

    ➔ Loss Function Mean Squared Error (MSE).
  - Reconstruction
    - Reconstruct the input X

    ➔ Loss function MSE.







Autoencoder neural networks

An autoencoder is a neural network trained to reconstruct its input.

Wind speed

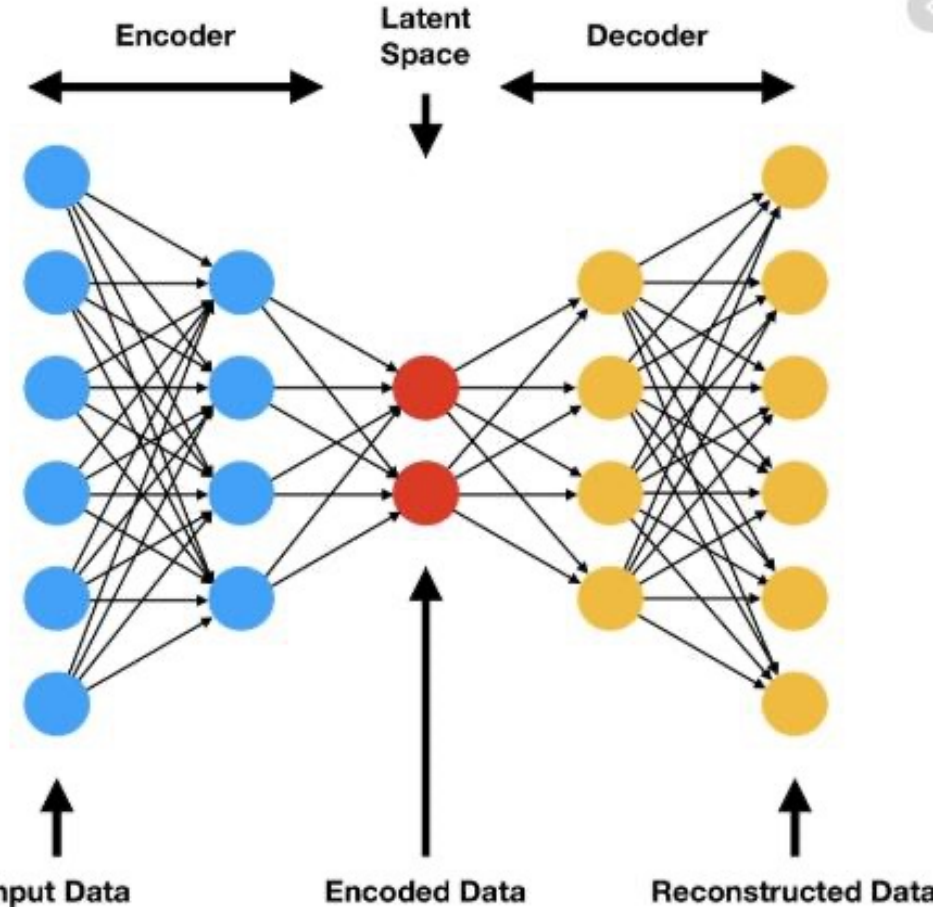
Ambient temp.

Rotor speed

power

Gen. Bear Temp.

Gear. Oil Temp.



Residual = reconstruction error:

$$r_j = X_j - \hat{X}_j$$

↓

Anomaly score

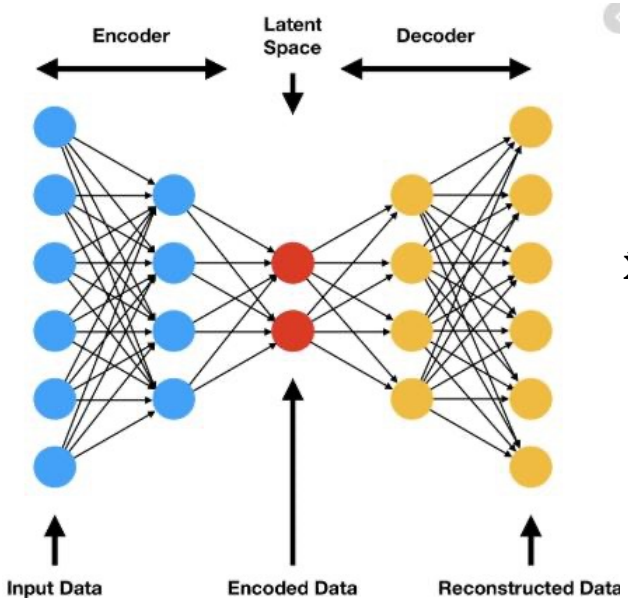
Z Yang, et al (2021)."Autoencoder-based representation learning and its application in intelligent fault diagnosis: A review

Mathematical formulation: reconstruction

- Input vector $\mathbf{x} \in \mathbb{R}^d$,
- Training set $\mathcal{D} = \{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(N)}\}$.
- Prediction task $\hat{\mathbf{x}} = f_\theta(\mathbf{x})$

→ Minimize. $\mathcal{L}_{rec} = \| \mathbf{x} - \hat{\mathbf{x}} \|_2^2 = \| \mathbf{x} - f_\theta(\mathbf{x}) \|_2^2$ over the training set → find θ^*

$\mathbf{x} \in \mathbb{R}^d$



$\hat{\mathbf{x}} \in \mathbb{R}^d$

Novelty detection methods

Common situation: only normal data is available.

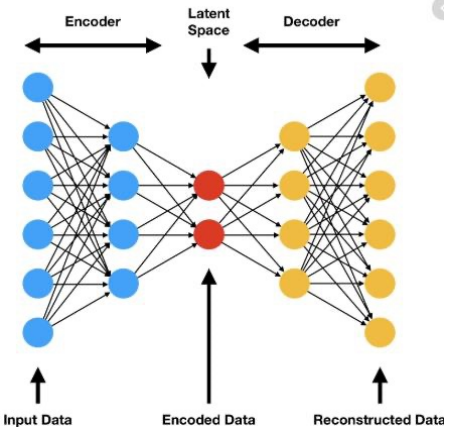
- Anomalies are rare/ costly/ risky.
- Not all expected types of anomalies have been observed.

➔Residual based approaches:

- train with clean (normal) data.
- Use residuals as anomaly scores.

Examples:

- Principal Component Analysis.
- AutoEncoder neural networks.
- Regression neural networks.



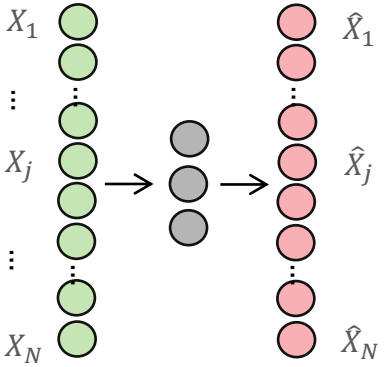
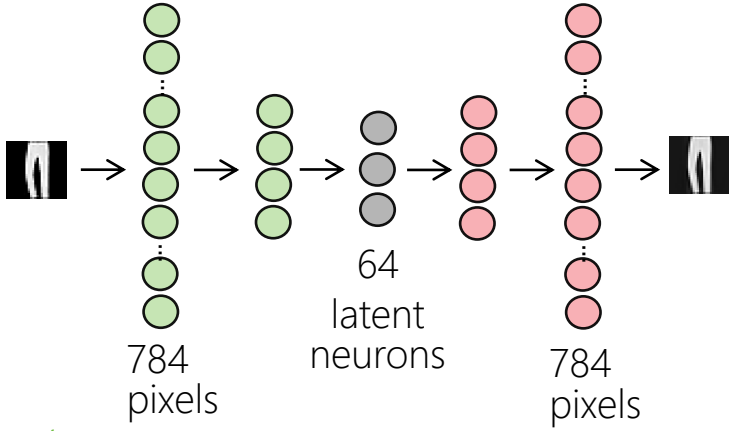
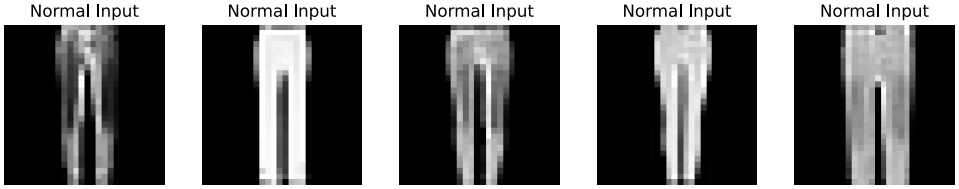


Image Anomaly Detection with an Autoencoder

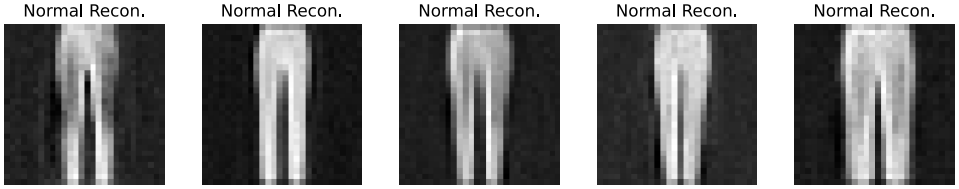
Example: trousers vs. all



Normal images

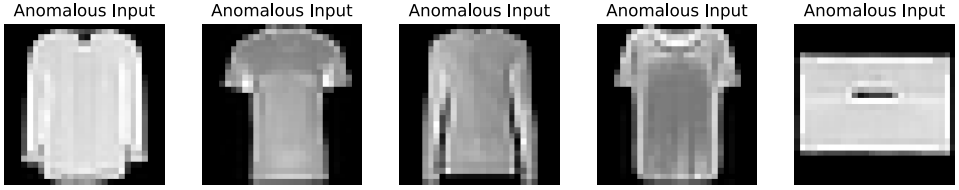


input

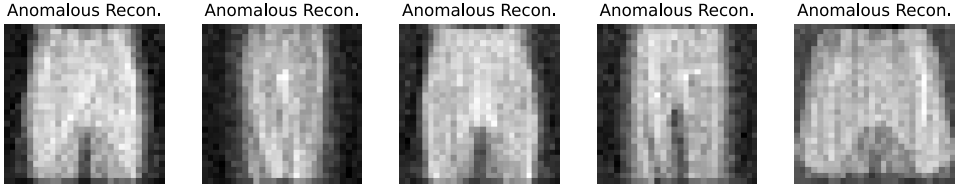


reconstruction

Abnormal images



input

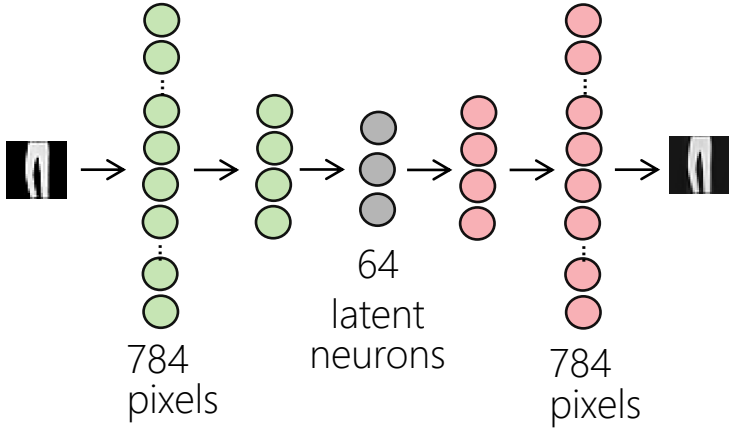


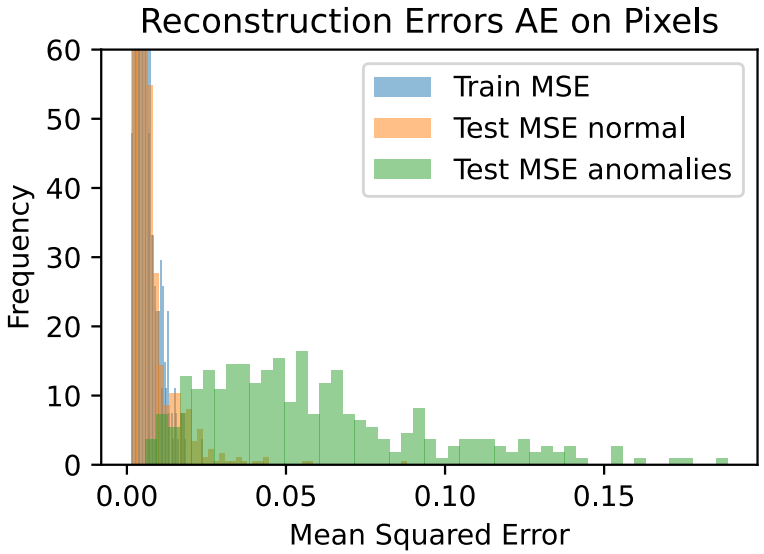
reconstruction

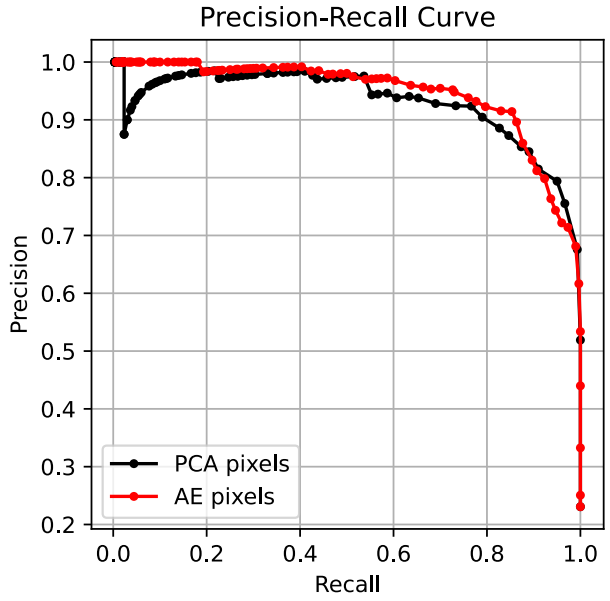
Image Anomaly Detection with an Autoencoder

- Example: trousers vs. all
- Latent dimension=64.
- Calculate the mean squared reconstruction error (MSE) for each image.

$$MSE = \frac{1}{N} \sum_{i=1}^{N=784} \left\| X_i - \hat{X}_i \right\|^2$$







WPI_AnoDet_05c_PCA_AE_FMNIST

Novelty detection methods

Only normal data available for training → residual based approaches

1. Train with normal (healthy) data.
2. Use residuals to derive anomaly scores.
3. Select a threshold to detect abnormal samples.

Works with

- Tabular data
- Time series
- images

